

UNDERSTANDING E-RUPI AND ITS IMPLICATION ON AGRICULTURE

A project report Submitted by

APOORVA

PGDM-ABM-2108

in partial fulfillment for the award of the degree

Post Graduate Diploma in Management (Agribusiness Management)

under the supervision of

DR. A. BHANDAPANI

**Head, Information and Communication Management
ICAR-NAARM**



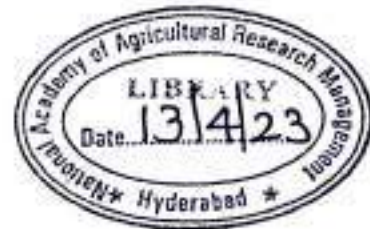
भारत-अनुष-राष्ट्रीय कृषि अनुसंधान आयोग
राजेन्द्रनगर, हैदराबाद-500030, तेलंगणा, भारत
ICAR National Academy of Agricultural Research Management
(ISO 9001:2008 Certified)
Rajendranagar, Hyderabad-500030, Telangana, India



APRIL 2023

-OR REFERENCE ONLY

Not to be Issued



UNDERSTANDING E-RUPI AND ITS IMPLICATION ON AGRICULTURE

A project report Submitted by

APOORVA

PGDM-ABM-2108

in partial fulfilment for the award of the degree

Post Graduate Diploma in Management (Agribusiness Management)

under the supervision of

DR. A DHANDAPANI

Head, Information and Communication Management

ICAR-NAARM



भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
राजेन्द्रनगर, हैदराबाद-500030, तेलंगाणा, भारत
ICAR-National Academy of Agricultural Research Management
(ISO 9001:2008 Certified)
Rajendranagar, Hyderabad-500030, Telangana, India



APRIL 2023

UNIVERSITÄT WÜRZBURG
FACHBEREICH AGRARWISSENSCHAFTEN

PG-420 / 13.4.2023

Prüfungsausschuss

AGRARWISSENSCHAFTEN

PG-420-2108

Prüfungsausschuss für die Agrarwissenschaften

Prüfungsausschuss für die Agrarwissenschaften (Agrarwissenschaften)

Prüfungsausschuss

AGRARWISSENSCHAFTEN

Prüfungsausschuss für die Agrarwissenschaften / Landwirtschaft

AGRARWISSENSCHAFTEN

Prüfungsausschuss für die Agrarwissenschaften
Prüfungsausschuss für die Agrarwissenschaften
Prüfungsausschuss für die Agrarwissenschaften
Prüfungsausschuss für die Agrarwissenschaften
Prüfungsausschuss für die Agrarwissenschaften



APRIL 2023



भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
राजेन्द्रनगर, हैदराबाद-500030, तेलंगाणा, भारत
ICAR-National Academy of Agricultural Research Management
(ISO 9001:2008 Certified)
Rajendranagar, Hyderabad-500030, Telangana, India



BONAFIDE CERTIFICATE

Certified that this project report " Understanding E-RUPI and Its Implication on Agriculture" is the bonafide work of "APOORVA" who carried out the project work under my guidance.

Signature

Dr. A DHANDAPANI

Head, Information and

Communication Management

ICAR-NAARM

Students Declaration

I do hereby solemnly and sincerely declare, to the best of my knowledge and belief that the project entitled **Understanding E-RUPI and Its Implication on Agriculture** is a bonafide record of work done by me during the course of the final project of the PGDM-ABM program under the guidance of Dr. A Dhandapani, Head, Information and Communication Management, ICAR-NAARM and all the contents and facts prepared and presented are authentic without any bias. The source of the materials used and all assistance received during the course of the investigation have been duly acknowledged.

I also declare that it has not previously formed the basis for any degree/ diploma/ Associateship or another similar title. The submitted report is my work and has not been duplicated from any other source.

Place: Hyderabad

Date: 10-April-23


APOORVA

PGDM-ABM-2108

Acknowledgment

My report would remain incomplete if I do not express my gratitude to the people who helped me out all the way through my final project. First, it is a matter and great privilege for me to be a part of the prestigious organization ICAR- National Academy of Agricultural Research Management, Hyderabad.

At the very outset, I extend a lion's share of my heartfelt gratitude, profound indebtedness, and veneration to **Dr. A Dhandapani** Head, Information and Communication Management, ICAR - NAARM) for excellent guidance, inspiring, unique, and compassionate steering of the challenging task and affection showing throughout the course of the project.

I take this opportunity to express my affection to my Parents & family for their consistent faith and support. I thank my batch mates of PGDM-ABM 2021-23 who were always beside me and supported me in completing my project. I want to thank all the respondents who have spent their valuable time giving authentic information for my project.

APOORVA

PGDM-ABM-2108

(ICAR-National Academy of Agricultural Research Management)

Executive Summary

The Reserve Bank, along with the Government of India, is in charge of the development, production, and overall administration of the country's currency with the aim of ensuring an adequate supply of genuine and clean notes throughout the economy. The changes in monetary history coexist with innovations in money and finance. The evolution of currency has included many different forms. India has come a long way in terms of innovation in digital payments. Technology development has enabled the creation of a new type of currency called bitcoin. The digital currencies of central banks (E-RUPIs). The RBI has also been weighing the advantages and disadvantages of the introduction of E-RUPIs for some time. Currently, it is working toward a phased implementation strategy, going step by step through different stages of pilots before the final launch, and is also looking into use cases for the issuance of its own E-RUPI (Digital Rupee (e)), with little to no disruption to the financial system.

Banks have the ability to program agricultural credit so that it can only be used at outlets that sell inputs. The same is true for MSMEs, etc., which could resolve the issue of financial inclusion and fund diversion. This could assist in ensuring the end-use that banks constantly struggle with around the world. To keep the fundamental characteristics of a currency, however, the E-RUPI's programmability feature needs to be carefully examined. The transmission of monetary policy may also be affected in other ways.

Tokens might have an expiration date by which they would need to be used, ensuring consumption.

E-RUPI in the financial sector can bring many changes and can help reducing the cost incurred by government on various factors such as printing of physical currency i.e. Rs. 5000 Cr annually, spending on the subsidies and allocation of the grants to various institutes and organisations. Usage of the E-RUPI will help in reduction of corruption and increase the transparency in the financial system. E-RUPI will also help in reduction of the commissions by the third party like banks.

It is a revolutionize the payments in the country.

LIST OF TABLES

CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

1.2 Problem Statement

1.3 Objective

CHAPTER 2: LITERATURE REVIEW

CHAPTER 3: DESIGN OF THE STUDY

3.1 Methodology

3.2 Study Period

3.3 Data Collection

3.4 Sample and Selection of Data

3.5 Study Area and Demographics

3.6 Limitations of the Study

CHAPTER 4: OBSERVATIONS AND FINDINGS

4.1 Digital Ecosystem of India

4.2 Digital India

4.3 Digital Ecosystem of the World

4.4 Digital India - 2020

4.5 Digital India - 2020 - A Vision

4.6 Digital India - 2020 - A Vision

Table of content

STUDENTS DECLARATION	III
ACKNOWLEDGMENT	IV
EXECUTIVE SUMMARY	V
TABLE OF CONTENT	VII
LIST OF FIGURES	IX
LIST OF TABLES	IX
CHAPTER 1: INTRODUCTION.....	1
1.1 INTRODUCTION	1
1.2 PROBLEM(S) ADDRESSED	2
1.3 OBJECTIVES	2
CHAPTER 2: LITERATURE REVIEW	4
CHAPTER 3: DESIGN OF THE STUDY	6
3.1 MATERIALS & METHODS	6
3.2 STUDY PERIOD	6
3.3 COLLECTION OF DATA	6
3.4 NATURE AND SOURCES OF DATA	6
3.5 STUDY AREA AND METHODOLOGY	6
3.6 LIMITATIONS OF THE STUDY	7
CHAPTER 4: OBSERVATIONS AND FINDINGS	8
4.1 DIGITAL LANDSCAPE OF INDIA	8
4.2 ABOUT E-RUPI	10
4.3 WORLD OVERVIEW OF THE E-RUPI	12
4.4 INDIAN SCENARIO	15
4.5 DIGITALIZATION IN AGRICULTURE	24
4.6 PAYMENTS IN AGRICULTURE	26

4.7 IMPLICATIONS OF E-RUPI IN AGRICULTURE	27
---	----

CHAPTER 5: RESULTS AND DISCUSSIONS	32
---	-----------

5.1 RESPONSES RECEIVED	32
5.2 AWARENESS ABOUT E-RUPI	33
5.3 WORKING OF E-RUPI	34
5.4 BENEFITS OF E-RUPI	36
5.5 IS E-RUPI A COUNTERPART FOR CRYPTOCURRENCY	37
5.6 TRANSACTIONS PROMOTED VIA E-RUPI	39
5.7 E-RUPI IMPACT ON INDIAN AGRICULTURE	40
5.8 FARMERS ADOPTION OF E-RUPI	40
5.9 RIGHTFUL USE OF SUBSIDIES	42
5.10 AGRICULTURE COMPANIES ADOPTION OF E-RUPI	43
5.11 AGRIBUSINESS SECTOR TO BENEFIT MOST FROM E-RUPI	44
5.12 CURIOSITY ABOUT E-RUPI	45

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS	46
--	-----------

CONCLUSION	46
RECOMMENDATIONS	46

REFERENCES	48
-------------------------	-----------

ANNEXURE	49
-----------------------	-----------

List of Tables	49
Table 1	50
Table 2	51
Table 3	52

List of Figures

Figure 1 Features of E-RUPI	2
Figure 2 Development in payment system in India	9
Figure 3 Concept of E-RUPI.....	10
Figure 4 Preferred Mode of Payment & to receive Money (In Percent).....	17
Figure 5 Preferred Mode For Small Value Transaction (In Percent)	18
Figure 6 Adoption Of Digital Payments as per Awareness	18
Figure 7 Adoption of Digital Payments as per Education Level.....	19
Figure 8 Types of E-RUPI	20
Figure 9 Forms of E-RUPI.....	21
Figure 10 Working of Account Based and Token Based E-RUPI	22
Figure 11 Prepaid Voucher Flow Process for Distribution.....	23
Figure 12 Flow process of entities included	23
Figure 13 Models of Issuance of E-RUPI.....	24
Figure 14 Fertilizer Import of India.....	27
Figure 15 Awareness E-RUPI.....	32
Figure 16 Profiles of the respondents	32
Figure 17 Regional Awareness of E-RUPI	33
Figure 18 Percentage Awareness about E-RUPI among Gender	34
Figure 19 e-RUPI works.....	35
Figure 20 Regional opinion on working of e-RUPI	35
Figure 21 Occupational Awareness regarding the working of E-RUPI.....	36
Figure 22 Benefits of E-RUPI	36
Figure 23 Regional Awareness of CE-RUPI.....	37
Figure 24 E-RUPI - A counterpart for Cryptocurrency	37
Figure 25 E-RUPI counterpart for Cryptocurrency (Region)	38
Figure 26 Will E-RUPI reap same benefits as cryptocurrency.....	38
Figure 27 Transaction done through E-RUPI.....	39
Figure 28 Transactions promoted through E-RUPI(Region)	39
Figure 29 E-RUPI impact on Agriculture	40
Figure 30 Will E-RUPI help Farmers.....	41
Figure 31 Farmers adoption of E-RUPI(Region)	41
Figure 32 Rightful use of Subsidies.....	42
Figure 33 Rightful Use of Subsidies(Region)	42
Figure 34 Agri- Companies Adoption of E-RUPI(Gender)	43
Figure 35 Readiness of Agri- Companies to Adopt E-RUPI	44
Figure 36 Agri- Sector Benefit most from E-RUPI.....	44
Figure 37 Curiosity about E-RUPI	45

List of Tables

Table 1 RBI DPI Index.....	16
Table 2 Comparison of Payment Methods.....	19
Table 3 Banks issue E-RUPI.....	24

Chapter 1: Introduction

1.1 Introduction

The changes in monetary history coexist with innovations in money and finance. Currency is a type of money that is issued only by the sovereign (or a central bank acting as its representative) and is considered legal tender in countries. India has come a long way in terms of innovation in digital payments. The payment ecosystem in India has been able to expand in a controlled manner. Advancements in technology have made it possible the development of a new form of money.

Right now, more than 70% of the world's central banks are exploring the merits of central bank digital currencies including India. Central Bank Digital Currencies as defined by Reserve Bank, named as E-RUPIs in India, broadly the legal tender issued by a central bank in a digital form. It is akin to sovereign paper currency but takes a different form, exchangeable at par with the existing currency and shall be accepted as a medium of payment, legal tender and a safe store of value. The use of the offline feature in E-RUPI would also be beneficial in remote locations and offer availability and resilience benefits when electrical power or a mobile network is not available.

CDBC is the digital means of currency which provides the benefits of a block chain in the system without the associated risk with block chain. The broad objectives to be achieved by the introduction of E-RUPI using block chain and other technologies as a 'more efficient and cheaper currency management system'.

In agriculture, the main advantage of E-RUPI is the prevention of the misuse of subsidies/crop loans etc., For example, the bank offers crop loans/AG Loan to farmers at a subsidized interest rate and this is getting credited to the bank account or given in cash, this is being utilised in a wrong way i.e. using it for anything other than agriculture (in most cases). E-RUPI is programmed in such a way that it can be utilised for purchasing only to which it's intended. Many AgTechs are working on developing block chain technology in agriculture and hence using the potential of the technology in agriculture. Some AgriTech start-ups working in block chain are TraceX, Fasal, etc. It can help the business to trace the transactions of goods and services. It will also mitigate the risk of storing physical cash.

It can help the business to trace the transactions of goods and services. It will also mitigate the risk of storing physical cash. It can help the business to trace the transactions of goods and services. It will also mitigate the risk of storing physical cash.

1.1 Objective

- Understanding the e-RUPI (E-RUPI).

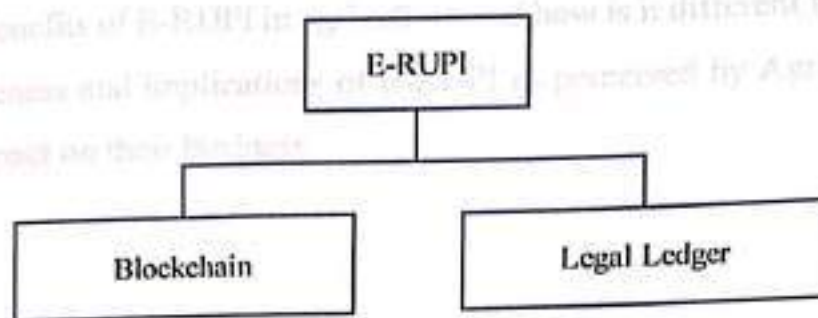


Figure 1 Features of E-RUPI

1.2 Problem(s) addressed

The government have provided subsidies of 4.03 lakh crores under 53 Ministries from 310 schemes to various sectors. Agriculture subsidies amount about to 36000 crores.

The agriculture subsidies are credited to the accounts of the farmers through DBT (Direct Benefit Transfer). There is no tractability of where the farmers utilize the subsidies given to them by the government. There is block chain technology is present in India with respect to the product and services given to the farmers but no tract of the subsidies given are being traced.

As subsidies will be directly credited to the wallet of the farmer, the impact on financial institutions can be seen. Due to lack of credit history availability, micro-finance companies deny credit and financial help to farmers due to which they shift to money lenders.

There are many financial intermediaries charging commissions from the farmer's transaction from one entity to another, E-RUPI will help in the direct transaction from the wallets of the payer to the payee through block chain technology without any interference from the financial intermediaries.

For the agri-inputs E-RUPI will help in tracing the farmers purchasing the goods and help in analysing the subsidies given by the government. The main aim of E-RUPI is to reduce the number of commissions taken by the banks through the currency exchange due to which the companies based in India have to pay more than the agreed price to the producer in order to fulfil the exchange commission rates charged by the banks. This can majorly be helpful in imports of fertilizers and the other imported commodities. Another aim of E-RUPI is to facilitate the transfer of international funds between companies and individuals

1.3 Objectives

- Understanding the e-RUPI(E-RUPI).

- The benefits of E-RUPI in agriculture and how is it different from digital money.
- Awareness and implications of E-RUPI as perceived by Agri-professionals and its impact on their business

H. Al *et al.* (2017) conducted a study regarding the E-RUPI with the perspective of the Monetary Policy. In this study, they compared the various method of payment such as cryptocurrencies, traditional banking and the E-RUPI. They also mentioned the various case studies of the E-RUPI being used by various countries. They also stated the various regulations that can be implemented in the E-RUPI. They also made a comparison in the physical and the digital method.

Sahil Dora and Shreshth Maheshwar (2023) conducted a study about the E-RUPI before the inception of E-RUPI in India. This article attempts to understand critical aspects of E-RUPI and its relevance for India, with a particular focus on rural E-RUPI. The shift from public fiat money to electronic and private money has opened a plethora of opportunities. At the same time, it has also posed challenges to the definition and traditional form of money, the role of central banks, and financial intermediation model. 40 percent central banks reported that they were actively engaged in research, experiments or development work related to E-RUPI. The survey indicated that 40 percent of central banks have progressed from research to experiments or proof-of-concept, whereas another 19 percent of respondent central banks have developed pilot projects. Financial inclusion, domestic payment efficiency, and payments safety are significant initiatives for central banks to consider E-RUPIs in emerging market economies.

Nikol Seifert and Patrick Doran (2021) conducted a study regarding the categories of the E-RUPI. They studied the categories and told us possible advantages related to Accessibility and Privacy. They also mentioned its potential disadvantages. In the study they concluded how different categories of the E-RUPI are being allowed and restricted.

Chapter 2: Literature Review

N. Ahmat and Sabrina Bashir (2017) conducted a study regarding the E-RUPI with the perspective of the Monetary Policy. In this study, they compared the various method of payments such as cryptocurrency, traditional banking and the E-RUPI. They also mentioned the various case studies of the E-RUPI being used by various counties. They also stated the various modulations that can implemented in the it. They also made a comparison in the physical and the digital method.

Sahil Deo and Sharadul Manurkar (2020) conducted a study about the E-RUPI before the execution of E-RUPI in India. This article attempts to understand crucial aspects of E-RUPI and its relevance for India, with a particular focus on retail E-RUPI. The shift from public fiat money to electronic and private money has opened a plethora of opportunities. At the same time, it has also posed challenges to the definition and traditional form of money, the role of central banks, and financial intermediation model. 80 percent central banks reported that they were actively engaged in research, experiments, or development work related to E-RUPI. The survey indicated that 40 percent of central banks have progressed from research to experiments or proof-of-concept, whereas another 10 percent of respondent central banks have developed pilot projects. Financial inclusion, domestic payment efficiency, and payments safety are significant motivations for central banks to consider E-RUPIs in emerging market economies.

Nikhil Sridhar and Patrick Horan (2021) conducted a study regarding the categories of the E-RUPI. They studied the categories and told its possible advantages related to Accessibility and Privacy; they also mentioned its potential disadvantages. In the study they concluded how different categories of the E-RUPI are being stored and transferred.

Anton N. Dikenko and Ross P. Buckley (2021) conducted a research on the E-RUPI working in the E-RUPI and the policy related to it. They also mentioned the major challenges pertaining to the financial services present in the Pacific islands. They made comparison among the E-RUPI around the world and suggested the next steps that can be taken for the development of the currency.

They based on demographic available in the public domain and a sample survey to elicit information from agricultural professionals. The respondents of the survey were from Pan India.

3.2 Study Period

The secondary and primary data was collected from various sources from the months of January to March 2023.

3.3 Collection of Data

Secondary data pertaining to the new technology usage was collected from RBI website and Agricultural Statistics at a Glance. The data pertaining to the potential usage of E-RUPI identified for the study were collected from the sources of government website. The data for GDP, savings and expense were taken from BIS website.

3.4 Nature and sources of Data

For the study, the data was collected from both primary and secondary sources. Primary data was collected from students, working professionals, etc. having an exposure to the agricultural business in the world and related to the finance industry.

3.5 Study area and methodology

The research is conducted at a Pan-India Level and the primary aim of the research is to frame the opinion of the Agri-Professionals working in the Financial sector and all other sectors of agriculture. For Collection of data both primary and Secondary data through

Chapter 3: Design of the Study

3.1 Materials & Methods

This chapter deals with the various aspects of methods and materials adopted for conducting the present study. This study is mostly based on documents available in the public domain and a sample survey to elicit information from agricultural professionals. The respondents of the survey were from Pan India.

3.2 Limitations of the Study

3.2 Study Period

The secondary and primary data was collected from various sources from the months of January to March 2023.

3.3 Collection of Data

Secondary data pertaining to the area, technology, usage was collected from RBI website and Agricultural Statistics at a Glance. The data pertaining to the potential usage of E-RUPI identified for this study were collected from the source of government website. The data for consumption and expense were taken from BIS website.

3.4 Nature and sources of Data

For the study the data was collected from both primary and secondary sources. Primary data was collected from students, working professionals, etc. having an exposure to the agriculture business in the world and related to the finance industry.

3.5 Study area and methodology

The research is conducted at a Pan –India Level and the primary aim of the research is to frame the opinion if the Agri-Professionals working in the Financial sector and all the other sectors of agriculture. For Collection of data both primary and Secondary data through

- Personal Interviews of managers working in microfinance and NBFCs and understanding their perception of E-rupee.
- Questionnaire through google forms to be circulated among the agri-professionals.
- Secondary data collected from various reports and articles by RBI, Harvard Business School, etc.

3.6 Limitations of the Study

- The study was conducted in a limited period of time, in a limited area of a particular socioeconomic situation and thus suffers from drawbacks and any generalization of the results could not be wholly unbiased.
- The findings of the study are based on the expressed opinions and recall mechanism of the respondents where the subjectivity might not have been completely overcome despite the best efforts of the investigator.
- The necessary primary data were collected from the Agri-Professional based on their memory by interview method and hence may have inherent impact on the accuracy of data.
- The data was collected digitally hence technical issues may hamper the accuracy of the data.

Chapter 4: Observations and Findings

4.1 Digital Landscape of India

India has come a long way in terms of innovation in electronic payments. The country's payment eco-system has been able to develop in an orderly fashion thanks to the passage of a separate law for Payment and Settlement Systems. The current cutting-edge payment systems, which are affordable, accessible, convenient, effective, safe, secure, and available 24 hours a day, seven days a week, 365 days a year, are a source of pride for the entire country because, in many cases, India is far ahead of the most developed nations in this area.

The Reserve Bank has announced the launch e- RUPI on August 2, 2021 India is now among the front-runners, as the majority of the world's major economies are yet to adopt the E-RUPI (E-RUPI). The startling change in payment preferences can be attributed to the development of reliable round-the-clock electronic payment systems like Real Time Gross Settlement (RTGS) and National Electronic Funds Transfer (NEFT), which have made seamless real-time or nearly real-time fund transfers possible. Additionally, the introduction of mobile-based payment systems like the Bharat Bill Payment System (BBPS) and National Electronic Toll Collection (NETC) to facilitate electronic toll payments have been pivotal developments that have changed the nation's payments ecosystem and garnered attention on a global scale. Because they offered consumers an alternative to using cash and paper for payments, the convenience of these payment systems ensured their quick acceptance.

The adoption of digital payments in the nation has been further strengthened by the inclusion of non-bank FinTech companies as PPI issuers, Bharat Bill Payment Operating Units (BBPOUs), and providers of third-party applications in the UPI platform. From commodities to metallic money to paper money to digital money, the idea of money has undergone significant evolution. The new financial landscape of the economy is being defined by the shifting characteristics of money. The digitalization of money is the next significant development in the history of money as a result of the development of cutting-edge technologies. Technology development has enabled the creation of a new kind of currency, called bitcoin. Cryptocurrencies issued by central banks (E-RUPIs).

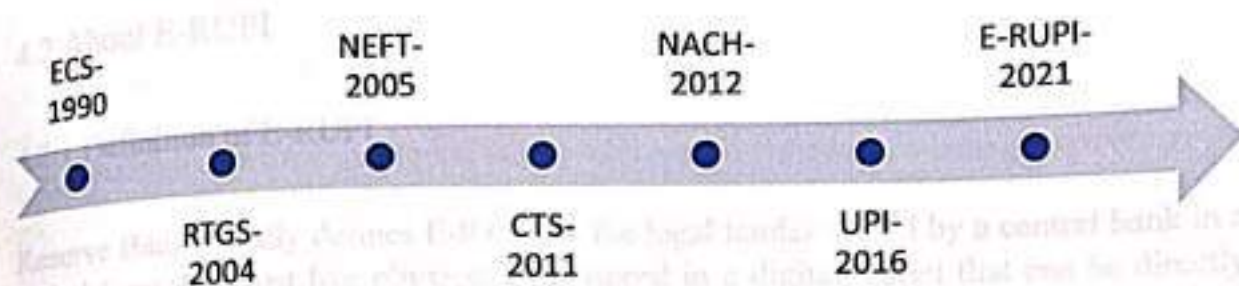


Figure 2 Development in payment system in India

With a development of the decentralized currency which claims the benefit of bypassing the regularised and monitored currency poses a threat to the already existing system hence to reap the benefit of the cryptocurrencies the central bank has come up with the digital currency.

The central bank provides either physical or digital currency through a channel of intermediaries such as financial institutions. To reduce the number of intermediaries between the final usage and the Central banks, it needed a solution through which legal tender which is a substitute for the physical cash can be exchanged between the entities digitally without the intervention of the financial institutions and the need for any internet connection.

These days, central banks issue two different kinds of currency and support a third with infrastructure. Reserves, also known as settlement balances or physical money, and electronic central bank deposits are issued. Peer-to-peer transactions using physical money are commonplace. In contrast, electronic reserves are held by central banks and typically only available to accredited financial institutions. Private money constitutes the third type of currency primarily accessible through deposits made to commercial banks using electronic means. Central banks. Support commercial bank money in various ways, including

- (i) Enabling commercial banks to settle interbank transactions Payments made in central bank currency;
- (ii) Allowing conversion between the central bank and commercial currency. Money is provided through the issuance of banknotes, and
- (iii) Contingent liquidity is provided through the lender of last resort function. Another significant point is that commercial bank deposits are not a liability of the central bank, only cash and reserves are not. A new kind of central bank currency would be E-RUPI.

4.2 About E-RUPI

4.2.1 Definition of E-RUPI

Reserve Bank broadly defines E-RUPI as the legal tender issued by a central bank in a digital form. It is just like physical cash stored in a digital wallet that can be directly transacted to the payee without the intervention of any institution or internet connection. It is also supported by the Central bank. It is legal tender issued by the Central Bank which can be transacted from one entity to other without it being stated invalid like other Wallets launched by the apps. A e wallet is a digital version of your physical wallet. The digital wallet, which is available on your Android device, will store your digital currency in the same way that physical bank notes are kept in physical wallets.

On August 2, 2021, the e-RUPI system, a one-time contactless and cashless payment method, was introduced. The Direct Benefit Transfer (DBT) scheme and digital transactions [Business to Business (B2B) transactions] among Micro, Small, and Medium-Sized Enterprises (MSMEs) will both be strengthened through the use of this digital platform. It is comparable to the forerunner of a E-RUPI (E-RUPI). In order to develop the "e-RUPI," the National Payments Corporation of India (NPCI) collaborated with the Department of Financial Services (DFS), the Ministry of Health and Family Welfare (MoHFW), the National Health Authority (NHA), and partner banks.

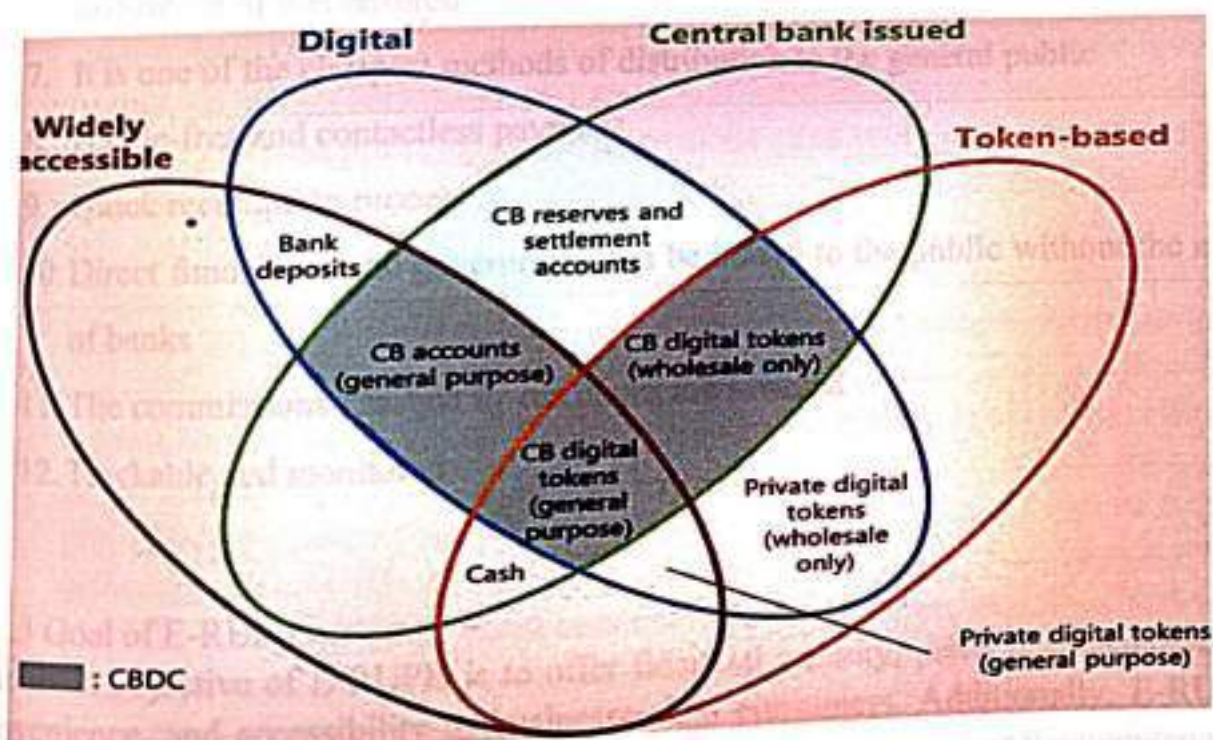


Figure 3 Concept of E-RUPI

e₹ wallet

As mentioned above the E-RUPI or e-RUPI will be stored in a digital wallet known as the e₹ wallet. An e₹ wallet is similar to your physical wallet in digital form. Available on your android device, the digital wallet will store your digital currency exactly as the physical bank currency notes are stored in the physical wallet. The money can be used for direct transactions from payer's wallet to the receiver's wallet without any interference of the banks. The e₹ wallet is just like a physical wallet but digitalised and can be used for transactions which are not completed using the path for the other digital payments. The money stored in the e₹ wallet is legal tender unlike the other wallets such as Amazon wallet, Bigbasket wallet, etc. and hence cannot be refused for making any transaction

4.2.2 Features of E-RUPI

1. Purely digital
2. Secure and faster transaction
3. No intervention of the intermediaries
4. Easy transaction
5. Can be used at any phone.
6. For Business use, any company can issue the vouchers through it and proper utilization of it is ensured
7. It is one of the cheapest methods of distribution to the general public
8. Hassle-free and contactless payment
9. Quick redemption process
10. Direct funds from the government can be issued to the public without the need of banks
11. The commissions charged by the banks are avoided
12. Trackable and monitored

4.2.3 Goal of E-RUPI

The main objective of E-RUPIs is to offer financial security, privacy, transferability, convenience, and accessibility to businesses and consumers. Additionally, E-RUPIs could lower the costs associated with cross-border transactions, lower the maintenance requirements of a complex financial system, and offer more affordable options to people who currently use alternative money transfer methods. Digital currencies issued by central banks would also lessen the dangers associated with using them as they are now.

Cryptocurrencies are extremely volatile, and their value changes all the time. A lot of households may experience severe financial strain as a result of this volatility, which could also have an impact on the economy's overall stability. E-RUPIs, backed by a government and governed by a central bank, would offer people's homes, businesses, and consumers a secure way to exchange digital currency.

4.2.4 Foundation Principles of E-RUPI

In their respective countries, central banks have a shared responsibility for monetary and financial stability. As a part of their public policy goals, institutions have been giving the public reliable money for centuries. The decisions they make regarding their public policies are a reflection of the current conditions and requirements in their jurisdiction. There are thus different and evolving policy options. However, there are three universal guiding ideas for a central bank's consideration of E-RUPI issuance that results from their shared goals.

1. **Do no harm is the motto.** The central bank should keep supplying new money forms, the accomplishment of public policy goals and shouldn't hinder or interfere with a central bank's capacity to fulfil its obligation to maintain financial and monetary stability. For instance, a E-RUPI needs to keep and bolster the uniformity or "singleness" of a currency while still allowing the general public to use various forms of payment, interchangeably.
2. **Coexistence.** In order to uphold stability, central banks move cautiously when entering uncharted waters, new (E-RUPI) and existing (cash, reserves, or settlement accounts) forms of central bank money, robust private funds (such as business bank accounts) should coexist and complement one another, to aid in achieving public policy goals. The provision and encouragement of cash by central banks should continue if the general public has an efficient demand for it.
3. **Efficiency and Creativity.** Without continued innovation and competition to increase efficiency in a. Users may use other, less secure instruments or currencies in place of the jurisdiction's payment system. This is ultimately, potentially harming consumer welfare and the economy, as well as the stability of the money supply. The. Public agencies, particularly the central bank, and private agents make up the payments ecosystem. (For example, payment service providers and commercial banks). Public and private sectors both have a part to play, the provision of payment services to build a secure, effective, and open system, private economic actors, generally free to choose the method of payment they will use for their transactions.

4.3 World Overview of the E-RUPI

As per the reports, the global payments have increased by 11 % in 2021 given the challenges of the pandemic and the changing payment habit of people across the world. It was estimated that about that overall electronic payment transactions grew in 2021 at a 19 % annual rate, in line with. Pre pandemic growth rates. Cash usage fell by 15% in 2020, which is due to COVID-19. In 2021, physical stores began to reopen, but the anticipated cash rebound did not occur. The 1% usage increase shows that the vast majority of transactions that switched to electronic channels in 2020 have stayed electronic within.

The Czech Republic, Greece, and the European Union, for example, experienced the biggest drops in cash usage between 2019 and 2021 (15 percentage points and 12 percentage points, respectively). In 2021, all regions experienced robust revenue growth, with Asia-Pacific and EMEA seeing increases of 13 percent. Despite being constrained by credit cards, both Latin America and North America experienced still-robust 7 percent growth rates.

Private virtual currencies are in stark contrast to the traditional understanding of money. Since they don't have any inherent value, they are not commodities or claims on commodities. Private cryptocurrencies have quickly exploded in popularity in recent years, posing a threat to our understanding of money's fundamental principles. Cryptocurrencies are being hailed as an innovation that will usher in decentralized finance and upend the established financial system because they claim the advantages of decentralization. Cryptocurrencies are built from the ground up to avoid the established, regulated intermediation and control mechanisms that are essential to maintaining the integrity and stability of the monetary and financial eco-system.

The Reserve Bank of India has consistently highlighted various risks associated with cryptocurrencies as the guardian of the monetary policy framework and with the responsibility to ensure financial stability in the nation. Due to their detrimental effects on the financial sector, these digital assets threaten India's financial and macroeconomic stability. A wider spread of cryptocurrencies also has the potential to limit the ability of the country's monetary authorities to control its monetary system and set its monetary policy, which could seriously jeopardize its financial system's stability.

According to the findings of a survey on E-RUPIs conducted in 2021 by the Bank for International Settlements (BIS) among 81 central banks, 90% of central banks are working on E-RUPIs in some capacity, and more than half are currently developing them or conducting practical experiments.

More than 60 central banks have expressed interest in E-RUPI globally; a few implementations in the retail and wholesale categories are currently undergoing testing, and many more are developing, testing, or launching their own E-RUPI frameworks. 105 nations are investigating E-RUPI as of July 2022, according to this statistic. 95 percent of the world's Gross Domestic Product (GDP) is covered. A E-RUPI has been introduced in 10 nations; the most recent was Jamaica's JAMDEX, with the first being

the Bahamian Sand Dollar in 2020. In the pilot stage and preparing for potential launches are 17 other nations, including powerful economies like China and South Korea. Piloting was first attempted by a major economy, China.

By April 2020, there will be a E-RUPI, and by 2023, it plans to have the e-CNY widely used domestically. E-RUPIs are being viewed more and more as a revolutionary development and the next development in the evolution of sovereign currency.

4.3.1 Case studies of Countries adopting E-RUPI

E-RUPI ideas have already been tested by the central banks of some nations. Lessons that might be applicable in other parts of the world are suggested by the experiences of Nigeria, China, and Jamaica.

Nigerian eNaira.

With the launch of retail E-RUPI eNaira in October 2021, Nigeria became the first country in Africa to adopt a digital currency. Its intended advantages include more efficient cross-border transaction capabilities, traceability and security to prevent fraud, and faster and more equitable distribution of cash assistance to households and communities taking part in social welfare programs. Over 800,000 people downloaded the eNaira app in the first seven months after it was released.

Some reports claim that only half of those downloads have been activated. Less than 100 active retailers were actively accepting payments in eNaira as of May 2022, which is a small number given Nigeria's status as the largest economy in Africa. Merchant adoption of digital currency has also been relatively slow. The low initial uptake of eNaira has been attributed to a lack of understanding of the E-RUPI and how it operates, worry about security breaches, and spotty internet access in some areas. The government of Nigeria recently declared that in response to these difficulties.

Unstructured Supplementary Service Data (USSD), which will be used to make eNaira available on feature phones, will increase the potential market by 100 million people in addition to the 25 to 40 million people who currently own smartphones. ¹ To raise awareness and identify important feature and technological advancements, the government recently sponsored a hackathon.

The digital yuan of China.

A large-scale E-CNY pilot spanning 15 cities was launched by the People's Bank of China, the country's central bank, in 2019. The E-CNY pilot, which focuses on transportation, government services, shopping, and other consumer-lifestyle use cases, has processed 260 million transactions totalling over 83 billion renminbi and 4.5 million merchant wallets as of May 2022. A fully functional E-CNY could address the last mile in light of China's comparatively high consumer electronic payment penetration converting China's payment system to a fully electronic, real-time one. E-CNY may

offer a substitute for international trade settlement, which is still heavily dependent on US dollars and the SWIFT system.

E-CNY uses a hybrid design model where the wholesale layer is account based and the retail layer is token based. According to publicly available documents, state-owned banks, commercial banks, and payments networks will all play operating roles, with commercial banks creating and maintaining both individual and merchant wallets. Even though the pilot has covered a lot more volume than any other country's E-RUPI program, it still represents a very small portion of China's total payments activity. There is still no official date or time set for the high-profile pilot expansion to Beijing in 2022 that is E-CNY. The absence of spectators dampened the impact of the Summer Olympics. Cross-border payments are currently being tested as a pilot project. Mainland-to-Mainland cross-border payment testing, for instance. A relationship between China and Hong Kong—which has a distinct legal system, financial system, and infrastructure—is currently underway. The Bank of China (Hong Kong), the second-largest commercial bank in Hong Kong, is a subsidiary of the state-owned Bank of China, and the pilot involves 200 of its employees as well as a select group of its merchant clients.

Jam-Dex from Jamaica.

Jam-Dex is a relatively straightforward retail offering with "streamlined" KYC requirements that went live in June 2022 and became the first E-RUPI to be officially recognized as legal tender. Its initial iteration of the product did not include advanced use cases like cross-border payments or smart contracts. Jam-Dex differs from the Bahamas' Sand Dollar and the Eastern Caribbean Central Bank's DCash because, despite utilizing distributed technology, it is not based on a block chain. Through partnerships with the private sector for interfaces and the issuance of digital wallets, the Jamaican Central Bank is pursuing an indirect model the ledger, infrastructure, and back end are all directly managed. The objective is to provide a seamless, safe, and user-friendly digital cash substitute. Early Jam-Dex use cases place a focus on peer-to-peer payments and payments to small and micro businesses, including those without conventional bank accounts, in order to increase financial inclusion.

4.4 Indian Scenario

Cost of cash management in India has continued to be significant. The total expenditure incurred on security printing during April 1, 2021 to March 31, 2022 was ₹4,984.80 crores as against ₹4,012.10 crores in the previous year. The growing use of the electronic medium of payment has not yet resulted in a reduction in the demand for cash.

In terms of volume, the payment systems saw a significant increase of 63.6 percent in 2021-2022 on top of the growth of 26.7 percent the year before. In terms of value, the

growth was 23% as opposed to a decline of 13% the year before, primarily because of the strong growth seen in the large value payment system. As a percentage of the total volume of non-cash retail payments, digital transactions made up 99.3% in 2021–2022 compared to 98.8% the year before.

The percentage increase in the value of banknotes during 2020-21 and 2021-22 was 16.8 per cent and 9.9 per cent respectively and the percentage increase in the volume of banknotes during 2020-21 and 2021-22 was 7.2 per cent and 5.0 per cent respectively. (Source: RBI Annual Report for the year 2021-22). The number of transactions using RTGS increased by 30.5 percent between 2021 and 2022 among digital payment methods. In terms of value, RTGS transactions registered an increase of 21.8 per cent; transactions through the National Electronic Funds Transfer (NEFT) system also witnessed an increase of 30.6 per cent and 14.3 per cent in volume and value, respectively, reflective of the increase in large value corporate transactions, in line with rising economic activity. NEFT services were available through 1,60,428 IFSCs of 227 member banks as of the end of March 2022, while RTGS services were offered through 1,56,740 IFSCs of 239 members.

Credit card payment transactions increased in volume and value by 27% and 54%, respectively, in the years 2021 to 2022. Debit card transactions decreased by 19% in terms of volume, but they increased by 10% in terms of value. Prepaid Payment Instruments (PPIs) saw increases in volume and value terms of 32% and 48%, respectively. The expansion of digital payments can be attributed to the acceptance infrastructure's increased accessibility, which grew significantly over the course of the year thanks to the PIDF's operationalization. During the year, the number of Points of Sale (PoS) terminals increased by 28.6% to 60.7% lakh, while the number of Bharat Quick Response (BQR) codes deployed increased by 39.3% to 49.7% lakh. Additionally, from 2.39 lakh the previous year to 2.48 lakh in 2021–22, there were more automated teller machines (ATMs).

Preferred mode of receiving money Debit or credit cards continue to be the most preferred mode of payment for 45% of the sample base Mobile wallets as an option is at No. 2 for 31% Only 9% preferred cash as a mode of payment However, for transactions more than Rs 10,000, 49% preferred debit or credit cards, 29% preferred net banking.

The increase in the the digital appetite of the people of the country suggests and motivates the Central banks to come up with new alternatives for digital transactions in the country The DPI (Digital payment Index) demonstrates the growth of the adoption and deepening of digital payments across the country.

Table 1 RBI DPI Index

Period	RBI-DPI Index
--------	---------------

March 2018 (Base)	100
March 2019	153.47
September 2019	173.49
March 2020	207.84
September 2020	217.74
March 2021	270.59
September 2021	304.06
March 2022	349.30

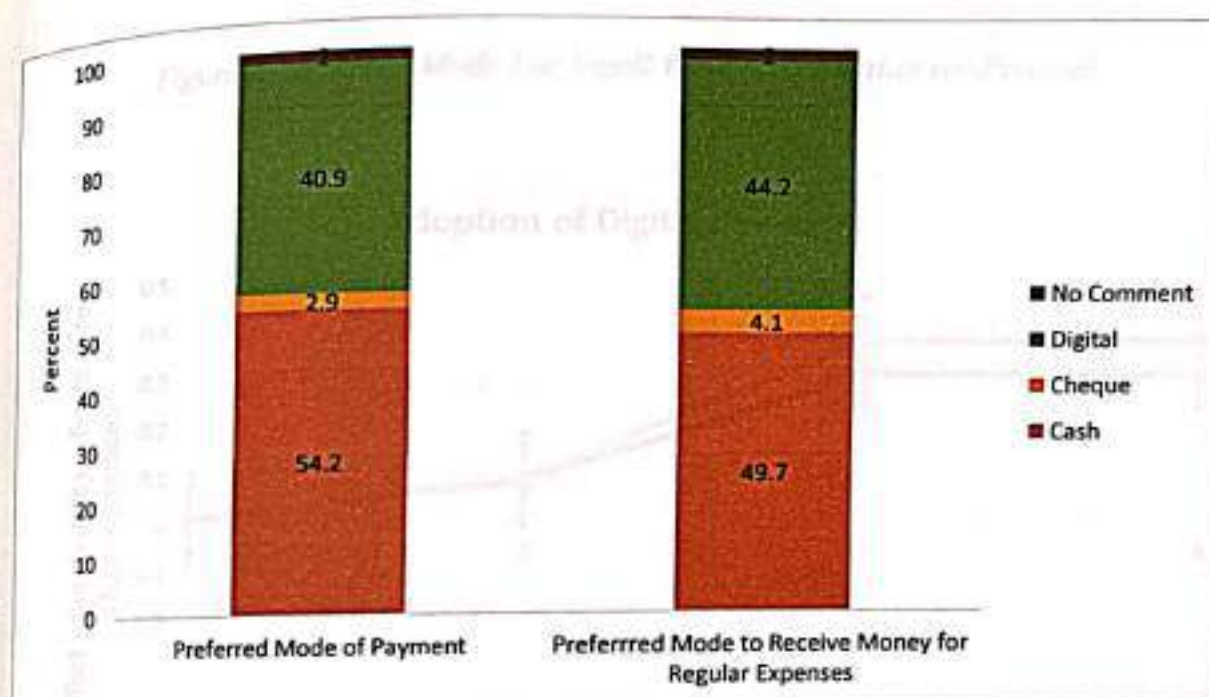


Figure 4 Preferred Mode of Payment & to receive Money (In Percent)

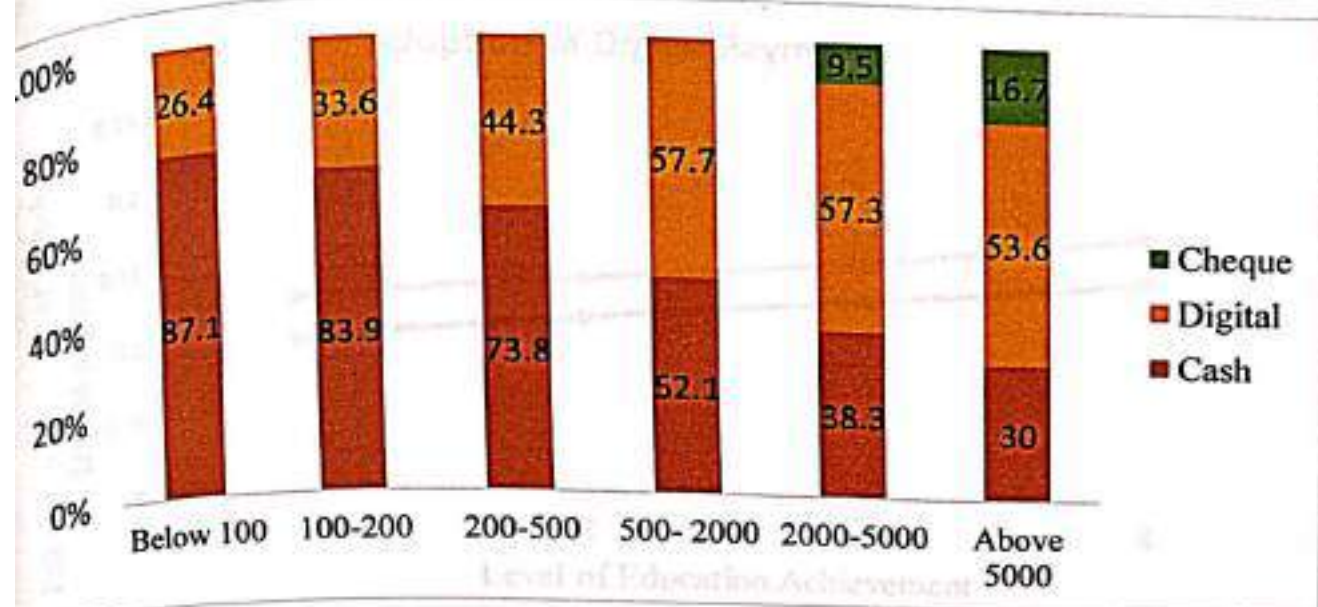


Figure 5 Preferred Mode For Small Value Transaction (In Percent)

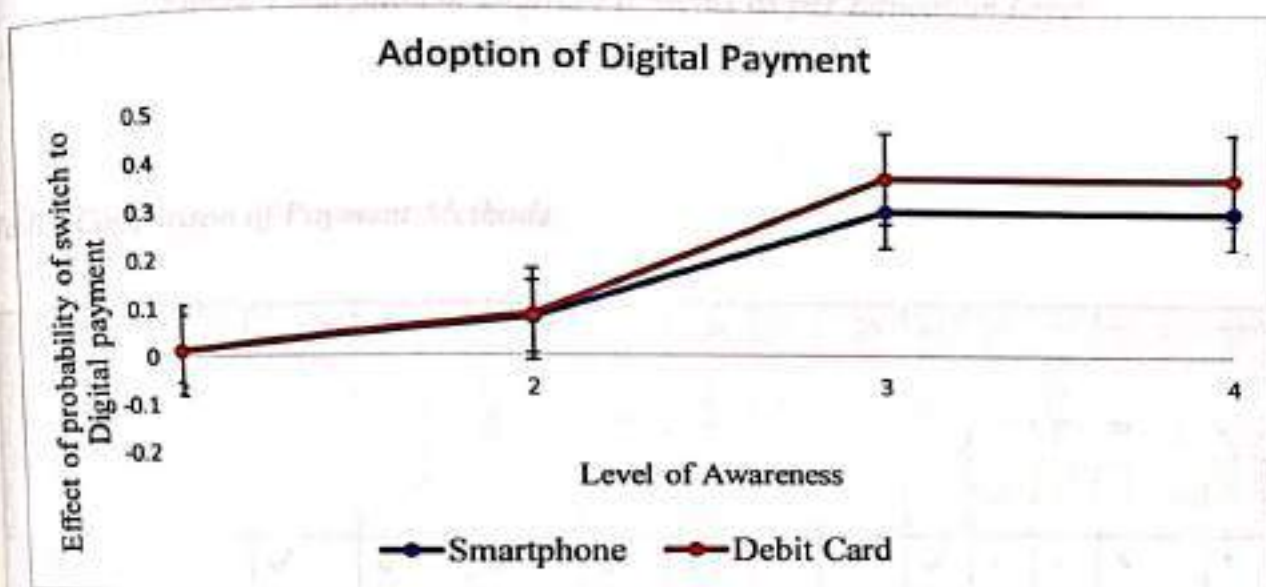


Figure 6 Adoption Of Digital Payments as per Awareness

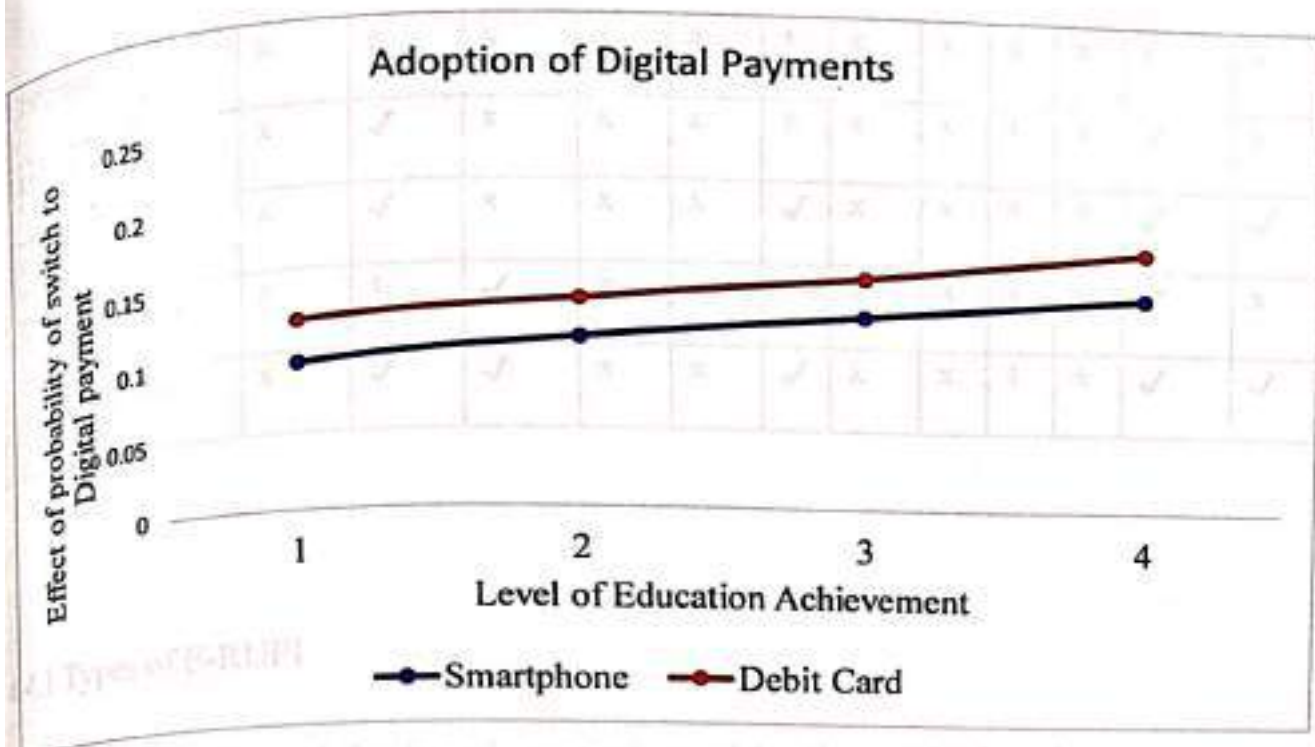


Figure 7 Adoption of Digital Payments as per Education Level

Table 2 Comparison of Payment Methods

	Legal Tender	Convenience	Direct Settlement	Anonymity	Availability	Reliability	Safe	Tangible	Secure	Efficient	Remote Payment	High Value
Cash	✓	✓	✓	✓	✓	✓	✓	✓	±	±	x	x
E-RUPI	✓	±	✓	x	x	✓	✓	±	±	✓	±	✓
Cards	x	✓	x	x	±	±	x	x	±	✓	✓	✓
Debit Card	x	✓	x	x	±	±	x	x	±	✓	±	±
Credit Cards	x	✓	x	x	±	±	x	x	±	±	✓	✓
Prepaid Cards	x	✓	x	✓	±	±	x	x	±	±	±	x
Credit Transfer	x	✓	x	x	±	✓	x	x	±	✓	✓	✓
Direct Debit	x	±	x	x	±	±	x	x	±	✓	✓	✓

Cheques	x	x	x	±	±	±	x	x	x	x	x	x
Mobile	x	✓	x	x	x	±	x	x	±	±	✓	x
Internets	x	✓	x	x	x	✓	x	x	±	±	✓	✓
Cryptocurrency	x	±	✓	±	x	x	x	x	x	±	✓	x
Instant Payment	x	✓	✓	x	x	✓	x	x	±	±	✓	✓

4.4.1 Types of E-RUPI

Based on the utility and the functions performed by the e-RUPI is demarcated into 2 categories i.e. General /Retail (E-RUPI-R) and Wholesale (E-RUPI-W)

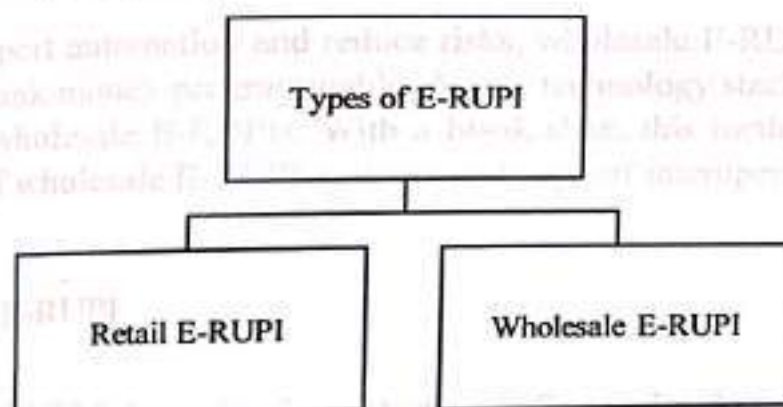


Figure 8 Types of E-RUPI

Retail E-RUPI (e₹-R)

It is the type of E-RUPI to be potentially used by the private sector a, non- financial consumers and businesses R-E-RUPI is an electronic form of cash primarily meant for the retail transactions as the name suggests. Retail E-RUPI can provide the access to safe money for the payment and settlement as it is a direct liability of the Central Bank. All the major transaction between the entities are to be done by the R-E-RUPI. It is the form of the type which is used for the day to day and business transactions between the entities.

The launch of E-RUPI-R will offer a secure central bank instrument with immediate access to central bank funds for settlement and payment. It is also asserted that it might increase a nation's retail payment systems' resilience. In the event that technical or operational issues disrupt the other payment system infrastructures, E-RUPI can offer a substitute method of making digital payments. E-RUPI could lessen the concentration of liquidity and credit risk in payment systems.

Wholesale E-RUPI (e₹-W)

This Type of the E-RUPI issued is different from the E-RUPI-R because it is more restricted to the institutions that can use it. The primary use of the E-RUPI-W is aimed to improve the efficiency of interbank payments or securities settlement. As per BIS Report the E-RUPI-W is intended for the settlement of the interbank transfers and related wholesale transactions. They are serving the same purpose as holding the reserves hold to the Central Banks but with an additional Functionality one such is the conditionality of payments, whereby the payment is only settled if the meet a certain condition. This may include many different conditional payment instructions, far more than the delivery-versus-payment mechanism used by real-time gross settlement (RTGS) systems at the moment.

In order to support automation and reduce risks, wholesale E-RUPIs could effectively make central bank money programmable. A new technology stack would also be used to implement wholesale E-RUPIs. With a blank slate, this method would enable the development of wholesale E-RUPI systems that support interoperability.

4.4.2 Forms of E-RUPI

The e-RUPI/E-RUPI is issued to the customers in 2 ways i.e. Account-based and Token based.

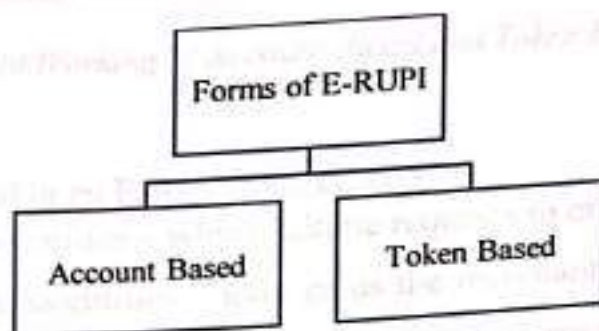


Figure 9 Forms of E-RUPI

Account Based E-RUPI
 An account-based approach typically uses a trusted third party to confirm a user's identity as the account holder and check their account balance prior to allowing them to make a payment. The appropriate debits and credits are then applied to the accounts. While the requirement for additional verification steps can repeat similar flaws with current legacy systems, this could result in unnecessary overheads.

Also impacted by the requirement for a third party is network governance. As a result, a wholesale E-RUPI (interbank settlements) seems better suited for an account-based approach because there is a clearer trade-off between accessibility and identity proof.

Token Based E-RUPI
 The central bank would issue a particular type of digital token that would serve as the digital equivalent of a banknote that could be electronically transferred from one holder to another. This digital token would represent a claim on the central bank. A token E-RUPI is a "bearer instrument," similar to banknotes, which means that whoever "holds" the tokens at any given time would be assumed to be their owner.

Account-based access compared with token-based access

Graph 4

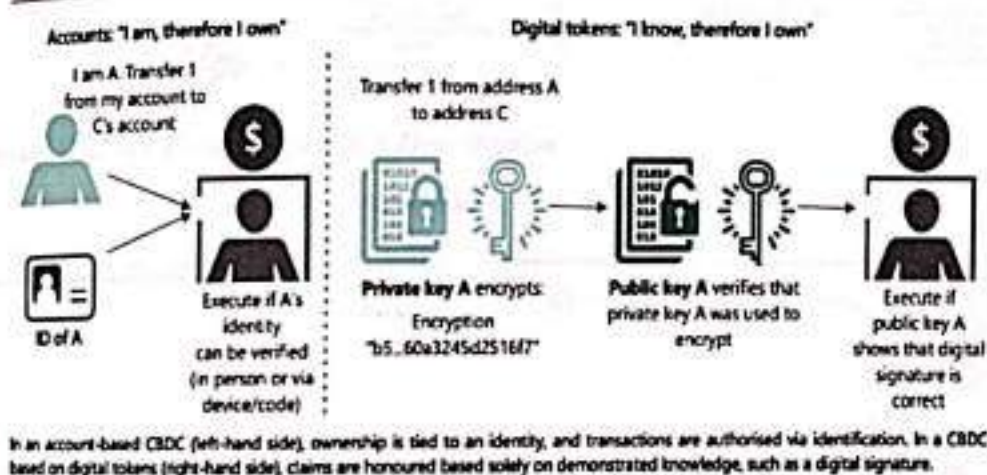


Figure 10 Working of Account Based and Token Based E-RUPI

4.4.3 Entities involved in an E-RUPI transaction

1. Issuers - banks/entities – which initiate requests to create UPI prepaid vouchers
2. Acquirers - banks/entities – gain gains the merchant settles the funds, acquirer banks/entities must be on board on the UPI platform to avail of e-RUPI

3. Sponsors – government, corporate or business customers of the bank who request for creation of UPI prepaid vouchers
4. Merchants – merchants/service providers will scan/read the QR/SMS the string from e-RUPI voucher and sends information to the on boarded acquirer entities for further processing
5. Beneficiaries - persons for whom the UPI prepaid voucher is issued. A beneficiary need not be a UPI user or have a bank account

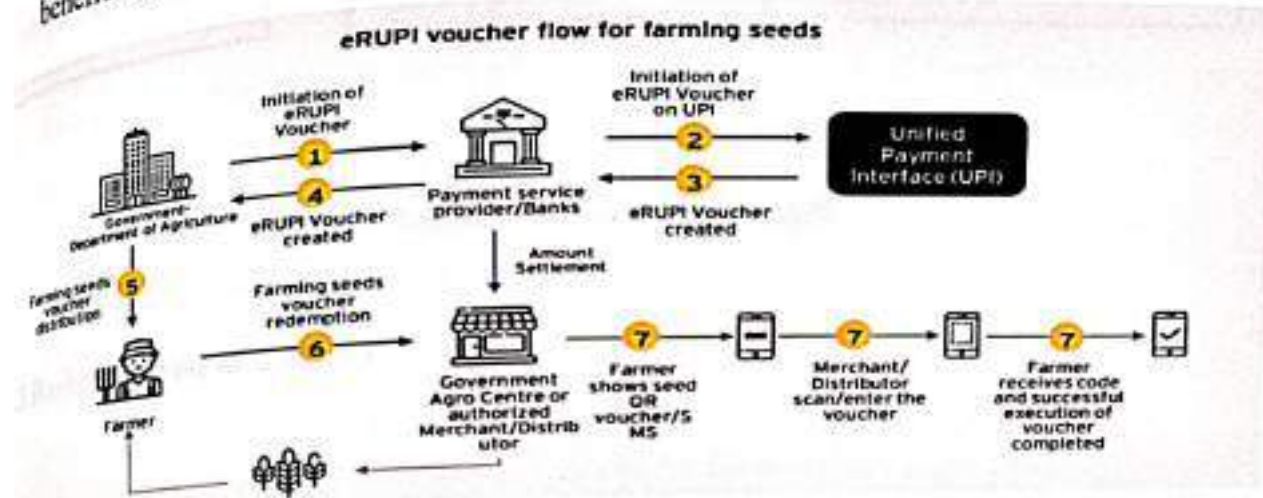


Figure 11 Prepaid Voucher Flow Process for Distribution

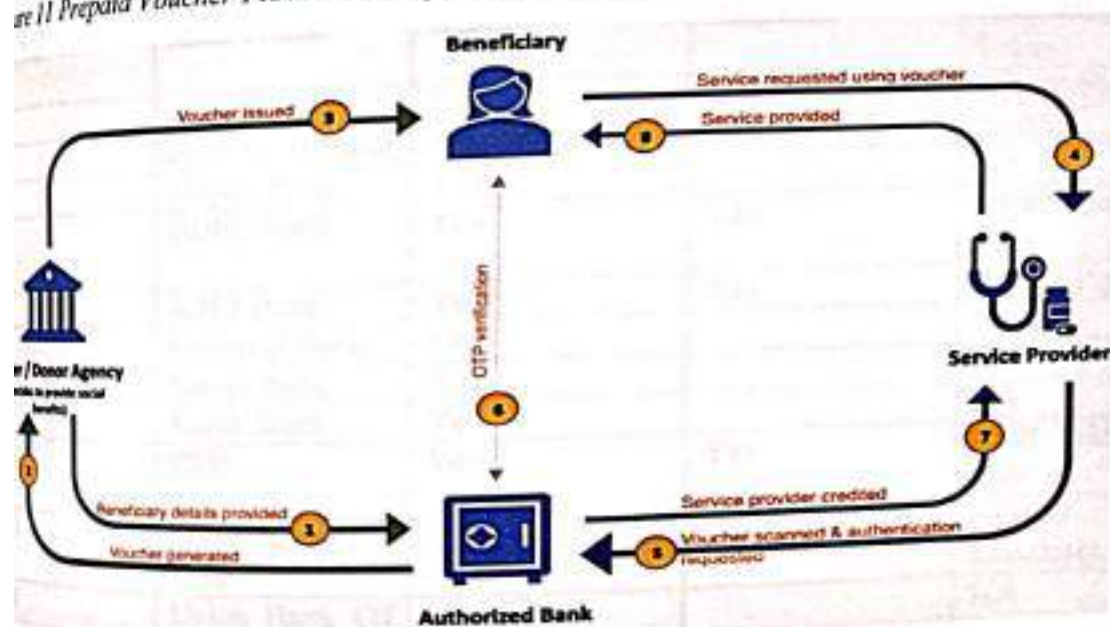


Figure 12 Flow process of entities included

4.4 Channels of Issuance of the E-RUPI

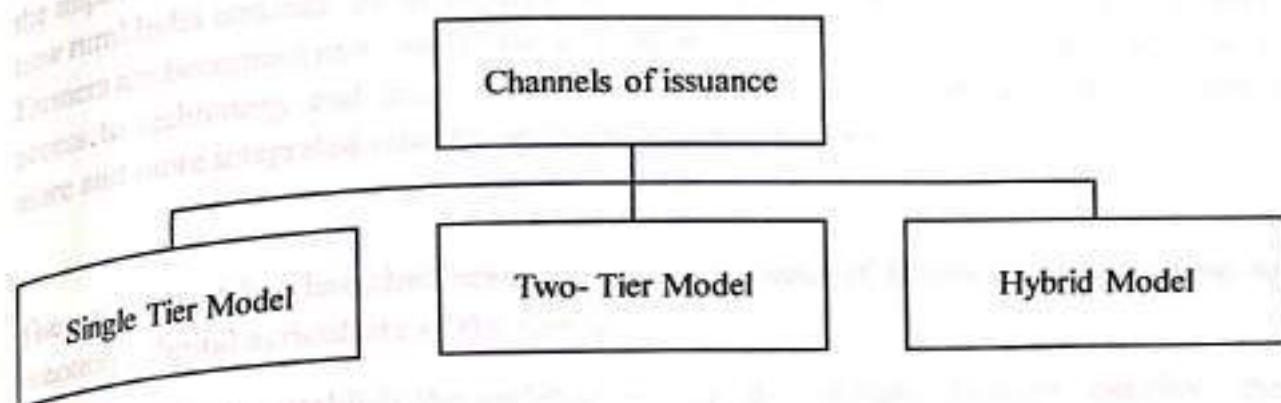


Figure 13 Models of Issuance of E-RUPI

4.4.5 Banks involved in E-RUPI

Table 3 Banks issue E-RUPI

Sr. No	Bank Name	Issuer	Acquirer	Acquiring App/ Entity
1	Axis Bank	Yes	Yes	Bharat Pay
2	Bank Of Baroda	Yes	Yes	BHIM
3	Canara Bank	Yes		NA
4	HDFC Bank	Yes	Yes	HDFC Business App
5	ICICI Bank	Yes	Yes	Bharat Pay
6	IndusInd Bank	Yes		NA
7	Indian Bank	Yes		NA
8	Kotak Bank	Yes		NA
9	PNB	Yes	Yes	PNB Merchant Pay
10	State Bank of India	Yes	Yes	YONO SBI Merchant
11	Union Bank Of India	Yes		NA

4.5 Digitalization in Agriculture

In the report by PIB, the Committee on Doubling Farmers' Income (DFI) recognized the importance of digital technology, which has the potential to fundamentally change how rural India conducts its agricultural activities by modernizing and organizing them. Farmers are becoming more informed as a result of various initiatives to give them easy access to technology and information, credit, and digital technologies are becoming more and more integrated into the agricultural value system.

The government has launched several initiatives, some of which are listed below, to encourage digital agriculture in the nation.

- In order to establish the architecture for the federated farmers' database, the government has finalized the core concept of the India Digital Ecosystem of Agriculture (IDEA) framework.
- In the plan scheme, i.e. Funds are released to the State(s)/UT(s) for projects involving the use of contemporary technologies, such as the National e-Governance Plan in Agriculture (NeGP-A), which uses modern technologies to implement agriculture. Robotics, drones, data analytics, block chain, artificial intelligence (AI), machine learning (ML), and other technologies.
- The Sub Mission on Agricultural Mechanization (SMAM) is currently being carried out. e. In April 2014
- National Agriculture Market (e-NAM) is a pan-India electronic trading portal which networks the existing Agricultural Produce Market Committee (APMC) mandis to create a unified national market for agricultural commodities
- Funds are directly transferred into the eligible farmers' bank accounts under the PM KISAN Scheme's Direct Benefit Transfer mode. Through the Farmers Corner in the portal, farmers can self-register. To increase the program's accessibility, the PM-KISAN Mobile App was released. Through it, farmers can check the status of their applications, update or correct their names in accordance with their Aadhaar cards, and view a history of bank account credits.
- Integrated Scheme for Agricultural Marketing schemes (AGMARKNET) to encourage the development of agricultural marketing infrastructure by providing backend subsidy support to State, cooperative, and private sector investments Services are provided through (AGMARKNET)
- vii. Agriculture Infrastructure Fund (AIF): To mobilize a medium-long term debt financing facility for investment in viable projects for post-harvest management Infrastructure and community farming assets through incentives and financial support in order to improve agriculture infrastructure in the nation. Beneficiaries like Farmers, Primary Agricultural Credit Societies (PACS), Farmer Producers Organizations (FPOs), Self Help Groups (SHG), State Agencies/APMCs receive

financial support digitally in the form of Interest Subvention and Credit Guarantee for setting up post-harvest management Infrastructure.

- The National Horticulture Mission encourages the sector's overall development, including the growth of bamboo. In order to achieve e-Government in NHM, a special intervention has been made in which complete transparency has been envisioned in every workflow process. e. application submission, authentication, processing, and payment to the beneficiary's bank account via DBT all take place online.
- To provide soil health cards to all farmers in the nation as a foundation for addressing nutrient deficiencies in fertilization practices as part of the national project on soil health and fertility. Farmers can follow soil sample progress on the Soil Health Card Portal.
- The creation of the mobile app Kisan Suvidha to make it easier for farmers to receive information on the crucial parameters, i.e. Weather, Market Costs, Plant Protection, Input Dealers (Seed, Pesticide, Fertilizer), Farm Equipment, Soil Health Card, and Cold Storages
- Additionally, the Indian Council of Agriculture Research (ICAR) has compiled and posted on its website more than 100 mobile applications created by ICAR, State Agricultural Universities, and Krishi Vigyan Kendras. These mobile apps, which were created in the fields of crops, horticulture, veterinary medicine, dairy, poultry, fisheries, and integrated subjects, provide farmers with useful information such as a set of best practices, market prices for different commodities, weather-related data, advisory services, etc.
- xii. The government sends SMS notifications to registered farmers with information on various crop-related matters.

4.6 Payments in Agriculture

Majorly the payments in agriculture were done in physical mode prior to the the Covid -29. The use of the physical cash has been reduced after Covid but people engaged in the farming sector still prefer to pay through the physical cash.

After the introduction of the the digital payment mediums such as NEFT and UPI people are getting more comfortable with the usage of them. To facilitate the Digitalization Movement in Agriculture the JAM Trinity was proposed i.e. Jan Dhan Account, Aadhar and Mobile. The main aim of the government was to propose the direct payments to the farmers without the misuse of the subsidies by the middlemen. This was also helpful in the Covid when major migration of the labours happened within the states, the money

was directly transferred to the accounts so that the people can directly use the government help.

In Agriculture the farmers are given the subsidies directly to their accounts through Direct Benefit Transfer. The subsidies were majorly given for to the farmers for the usage of the money to purchase the agriculture implements such as inputs, help in logistics, warehouse, greenhouse, etc. Though the money is directly transferred to the accounts of the farmers but the utility of the subsidy is not known very well.

The fertilizer has been a majorly imported commodity in agriculture. The import of Urea in the year 2021 has amounted to the value of more than 2500 million US dollars. The amount is generally paid by using the international methods such as SWIFT. Due to the changing difference between the currency rate the prize of the commodity fluctuates and hence the commodities are also charged with the exchange rates and commission accounted by the banks.

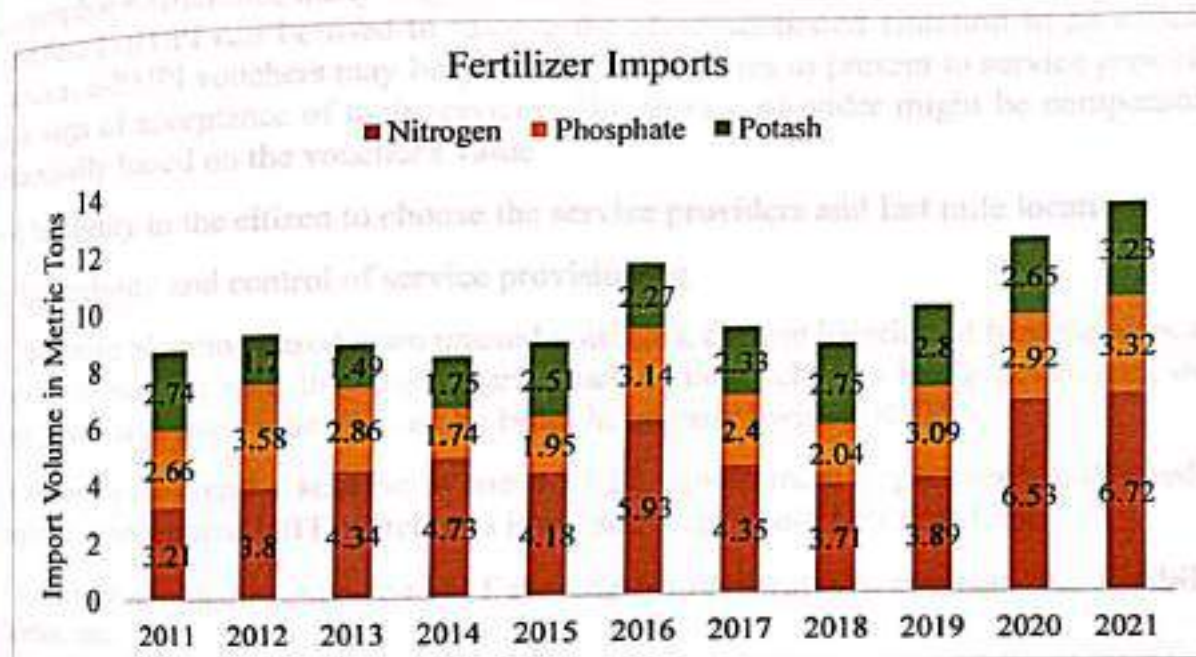


Figure 14 Fertilizer import of India

4.7 Implications of E-RUPI in Agriculture

The tracking of the funds allotted is a major drawback for the subsidies hence E-RUPI being a decentralized mode can be used for the transactions without the involvement of the intermediaries and the 3rd parties.

In agriculture the E-RUPI can help in various sectors such as Supply Chain as it will enable the faster and efficient transfer of the funds from payer to the Payee.

4.7.1 Direct Benefit Transfer

In India, Direct Benefit Transfer (DBT) programs can primarily be distributed using e-RUPI. These programs, which are a lifeline for the majority of them, have an impact on millions of people in India. Direct payments totalling more than 18 lakh crores are currently being made to beneficiaries nationwide who are linked to Aadhaar accounts across 311 schemes of 54 ministries. Between 40 and 60 percent of households in India are thought to use DBT in some capacity, with states like Uttar Pradesh and Bihar having even higher penetration rates. A few of the obstacles that e-RUPI might be able to help remove in order to increase the penetration of DBT include the lack of telecom adoption, ongoing population migration, and low levels of banking adoption among marginalized groups. Over 1000 non-DBT or in-kind programs exist in addition to the DBT-covered programs. The last-mile delivery, tracking, and feedback mechanisms for these programs—which make use of intermediary service providers—are extremely challenging to implement. Furthermore, due to migration, infrastructure problems, and geographic restrictions, many eligible beneficiaries are unable to utilize these non-DBT benefits. E-RUPI can be used to resolve the aforementioned situation in an efficient manner. e-RUPI vouchers may be given to beneficiaries to present to service providers as a sign of acceptance of their services. The service provider might be compensated financially based on the voucher's value

1. Flexibility to the citizen to choose the service providers and last mile locations
2. Traceability and control of service provisioning
3. Proactive planning based upon unused vouchers, current location of beneficiaries and providers Starting with the easy targets, such as new schemes being developed, over time, more and more schemes can be brought on board with e-RUPI.

In order to on-board a scheme to use e-RUPI, government departments will need to enhance government DBT systems to issue and acquire e-RUPI vouchers:

1. Link with the existing scheme DBT systems for registrations, applications, eligibility checks, etc.
2. Connect with SMS/MMS/email gateways for e-RUPI voucher distribution (text-only and QR Code)
3. Develop a system for bulk G2C pre-approved e-RUPI Voucher generation—linked to identity of eligible beneficiaries
4. Develop a scalable system and process to register and on-board service providers / merchants digitally for accepting e-RUPI
5. Work with banks for issuance and acceptance of e-RUPI vouchers

4.7.2 Sanctions on a Country

The usage of the E-RUPI in the international transaction will help to reduce the risk of the fluctuating currency rates and will enable the companies to make the transactions internationally efficiently and directly without the involvement of any 3rd party for making the transactions. Currently the system of SWIFT is being used to carry the transactions all over the world which is a system under the overseen by the 11 national banks of the G-10 – Belgium, Canada, France, Germany, Italy, Japan, the Netherlands, United Kingdom, United States, Switzerland, and Sweden and is jointly owned by thousands of financial institutions that use its system. As it is a system which is controlled by only 11 major countries, the collective decision of only 11 countries is powerful enough to cut-off a country from the facility of International Trade.

This specific case has been seen to be implied on Russia during the Russia – Ukraine war. It was collectively decided by the G10 countries to post sanctions against Russia in order to cut off any transactions that are being made. Due to the sanctions issued the country of Russia was not able to transact with other nations. E-RUPI allows the transactions internationally by bypassing the system of SWIFT and also help the data protection of the countries as the E-RUPI uses the Decentralized technology and no interference of the banks in the transactions.

4.7.3 Imports of Commodities

The commodities are generally transacted through a currency exchange in the respective 2 countries. For every transaction made the banks will charge a commission fees for performing the exchange of the currency. IN agriculture it is seen internationally that commodities purchased by a country is sorted and exported to the third country. It is done to classify the imported goods into the processed commodities and hence higher price in exports.

In India, Fertilizers are majorly purchased by China. The transactions made between the 2 companies in the 2 countries are charged by the banks and the third parties to carry out the exchange of the currency received. The fees can vary from 10- 12% per transactions. In year 2021 the value of import of Urea in India amounts to 2585 Million US Dollars hence only for the import of one commodity i.e Urea the commissions and fees charged by the banks to carry the transactions amounts to 258-310 Million US Dollars.

Hence due to the high exchange rates the companies have to pay more than 250 million to the banks and the third parties. To facilitate the companies in the import of the commodities without the involvement of the banks the companies in India can directly issue the E-RUPI and transfer it directly to the companies in the foreign land and there the companies can convert them into the currency and use it there. This helps in

reducing the exchange rates that are charged from the companies and the fees to carry an international transaction.

4.7.4 Tracking of the currency

The currency exchanged in the physical means and direct transfer of the funds loses its traceability and hence is used by the people for the uses other than the intended hence the E-RUPI can help to track the funds issued by the government

It can also help to reduce the chances of corruption and fondling of money from between. Tracking of the currency can also help to recognise the real consumers of the issued E-RUPI and can track the journey of the currency from one individual to another.

4.7.5 Buying Behaviour of the Individuals

E-RUPI can help to track the funds issued by the government to the individuals and also helps to understand the buying behaviour of the people in order to facilitate the subsidies in the right direction and reduce the wastage of the subsidies by the government on the things it is not intended to spend on. E-RUPI can also ease the usage of the funds by the farmers. Right usage of the subsidy is ensured.

4.7.6 Value Chain Transactions

There are many people involved in a supply chain or a value chain such as distributors, wholesalers, retailers, etc. The transactions are made at every step and the people take their share of portion and pay to the backend. E-RUPI can help to track the commission or charged taken at every step. It prevents the third parties to do money laundering in any manner and ensure the payments are made to the intended person.

4.7.7 Boost to the Start-ups

As the new form of the currency is Decentralised and directly under the control of the Central Bank hence the new startups can venture into the Competition to issue, acquire and utilize the E-RUPI. There has been increase in the digital payments made by the people in the recent times. The presence of the startups in the digital payment space will ensure new inventions and healthy competition among the digital payment space. The banks having their own app is already existing competition in the market such as YONO SBI App, HDFC Business App, PNB Merchant Pay, etc.

4.7.8 Utilization of the funds by the Institutes

Every government institute is given a grant by the government for better governing of the institute. The funds are mainly granted for the betterment of the infrastructure, towards student welfare, research and development of the crops and technology in

agriculture. The utilization of E-RUPI ensures that the funds are used for the intended use only and the diversion of the funds to unnecessary means can be tracked and abstained. Also by tracking the funds used by the institutes the government can decide the exact amount of funds required by the institute to prevent any wastage of the funds.

4.7.9 Cost of the goods

Using the E-RUPI technology the uniform cost of goods can be maintained all over the country. It ensures that the customers are not being charged more than the price set by the government by any trader or intermediary involved in the supply chain. It ensures fairness and fair price to each and every one involved in agriculture.

5.1 Responses received



Figure 13 Australian E-RUPI
Profile of Respondants



Figure 14 Profile of Respondents

Chapter 5: Results and Discussions

In accordance with the pre-determined objectives of the study, this chapter deals with the presentation and description of results emerged from the research work.

The primary survey was conducted among the Agri- Professionals. 102 responses within in the time period of 3 months i.e. January 2023- March 2023. The form was circulated through online means and a pan India study was held.

5.1 Responses received

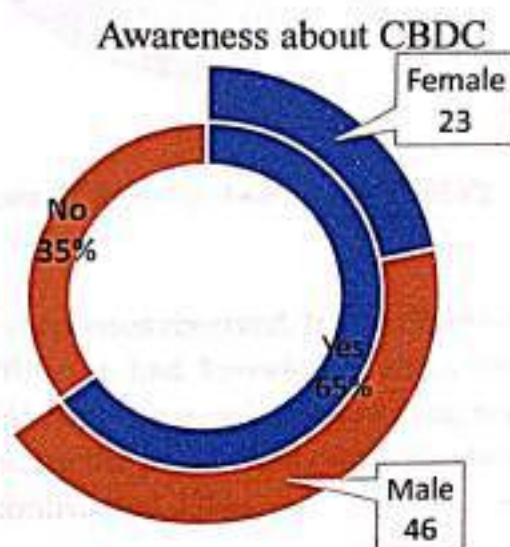


Figure 15 Awareness E-RUPI

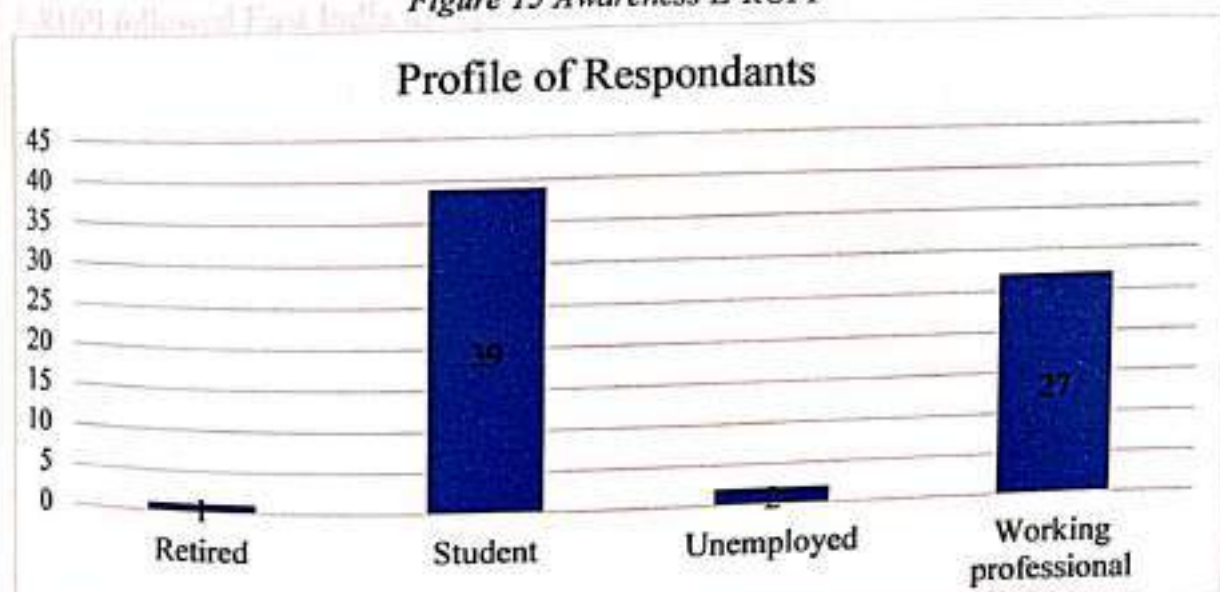


Figure 16 Profiles of the respondents

5.2 Awareness About E-RUPI

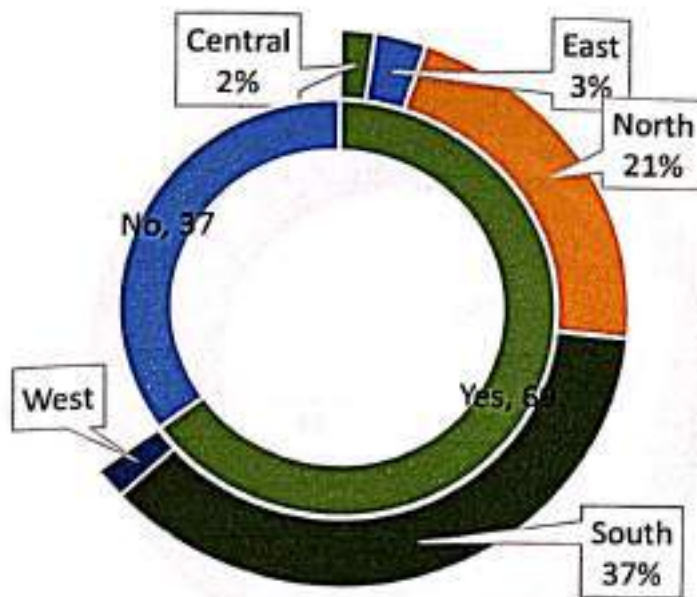


Figure 17 Regional Awareness of E-RUPI

The Figure 15 represents the responses received. It was observed that 65% of the people related to agriculture and finance had knowledge about the new digital currency technology i.e. E-RUPI and 35% of the people related to agriculture and finance didn't know about the E-RUPI. Among the people who were aware of the Males constituted about 66% and the females contributed 33%.

From the responses received, southern region of India contributed to about 37% of people aware of E-RUPI followed by North India contributing 21% of people aware of E-RUPI followed East India as represented in the Figure 17.

Percentage Awareness among Genders

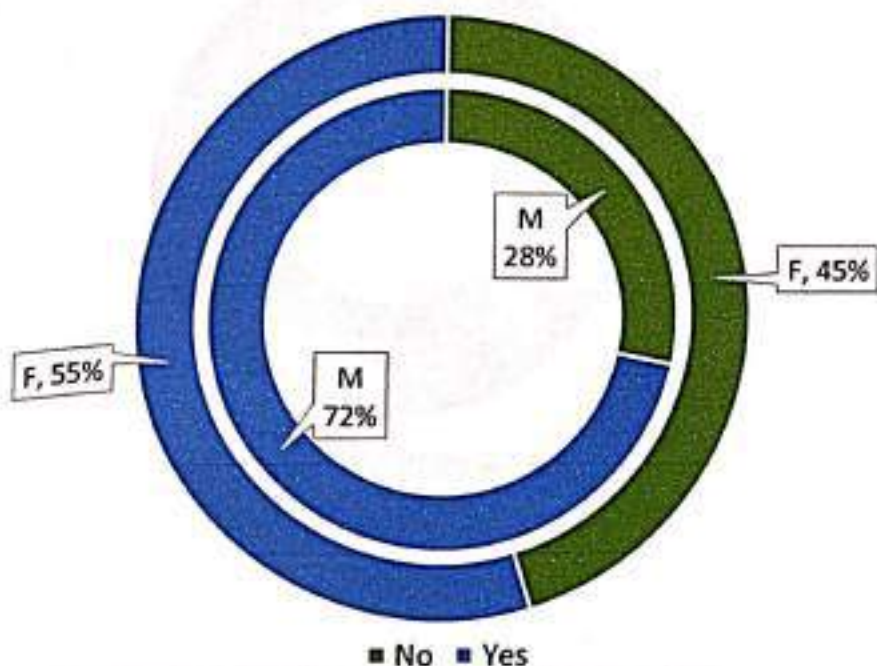


Figure 18 Percentage Awareness about E-RUPI among Gender

Figure 18 represents the relative comparison among the awareness among the genders. It was observed that 72% of the men were aware of the E-RUPI as compared to 55% of Females.

It was observed that the Central and Eastern India was more aware of the new digital technology and were aware of how this new method can revolutionize the Indian Agriculture. When compared in terms of the occupation, students were more aware.

5.3 Working of E-RUPI

How does e-RUPI works

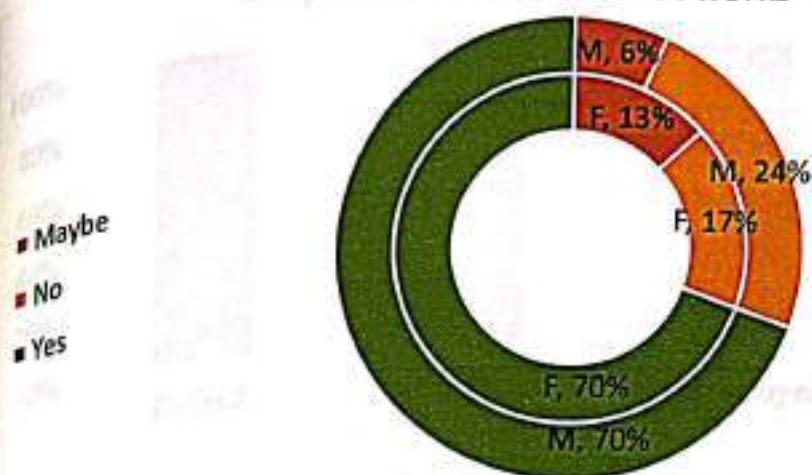


Figure 19 e-RUPI works

Are you aware about how e-RUPI works?

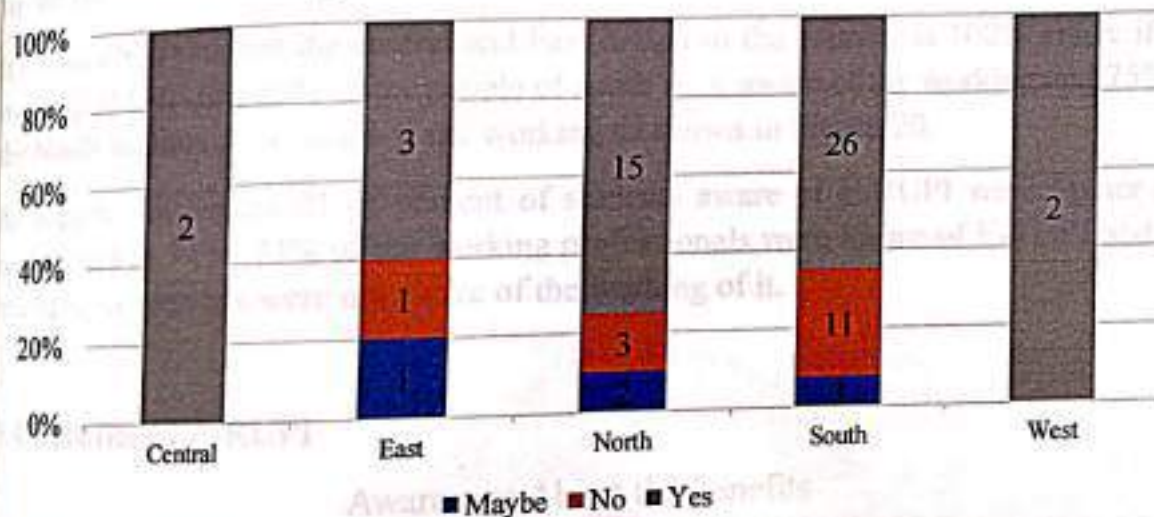


Figure 20 Regional opinion on working of e-RUPI

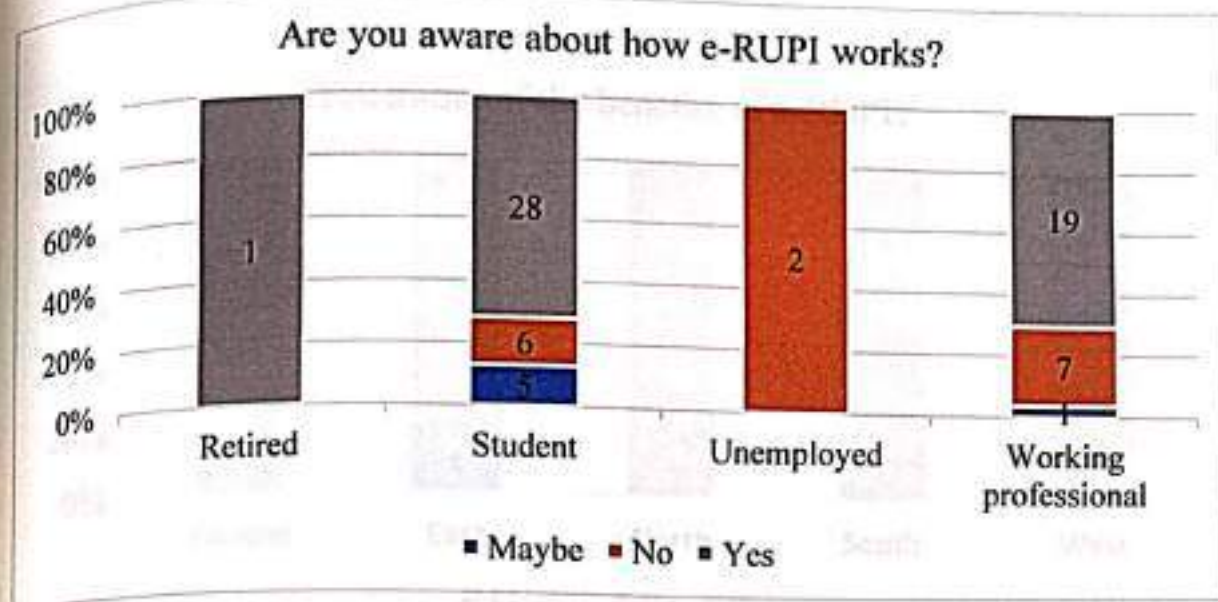


Figure 21 Occupational Awareness regarding the working of E-RUPI

Figure 23 Regional Awareness of E-RUPI

It was observed that 70 % of people were aware how E-RUPI works as depicted in figure 19.

It is also observed that the central and East region of the country is 100% aware if the working of E-RUPI. 65% of the people of south were aware of the working and 75% of the North Indians were aware if the working as shown in Figure 20.

As depicted in Figure 21 of percent of students aware of E-RUPI were higher and contributed to 71%. 74% of the working professionals were aware of E-RUPI and the unemployed persons were not aware of the working of it.

5.4 Benefits of E-RUPI

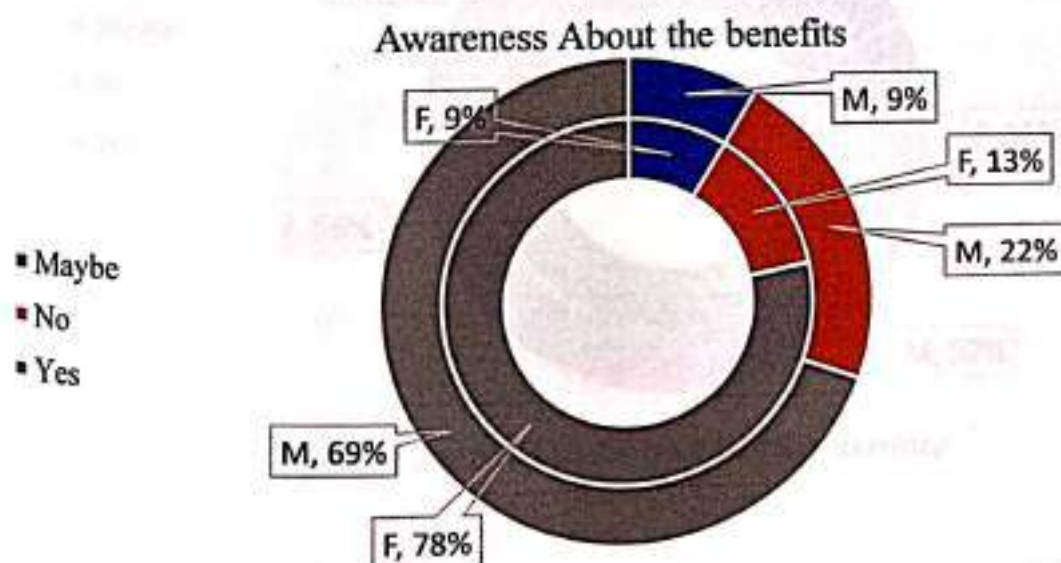


Figure 22 Benefits of E-RUPI

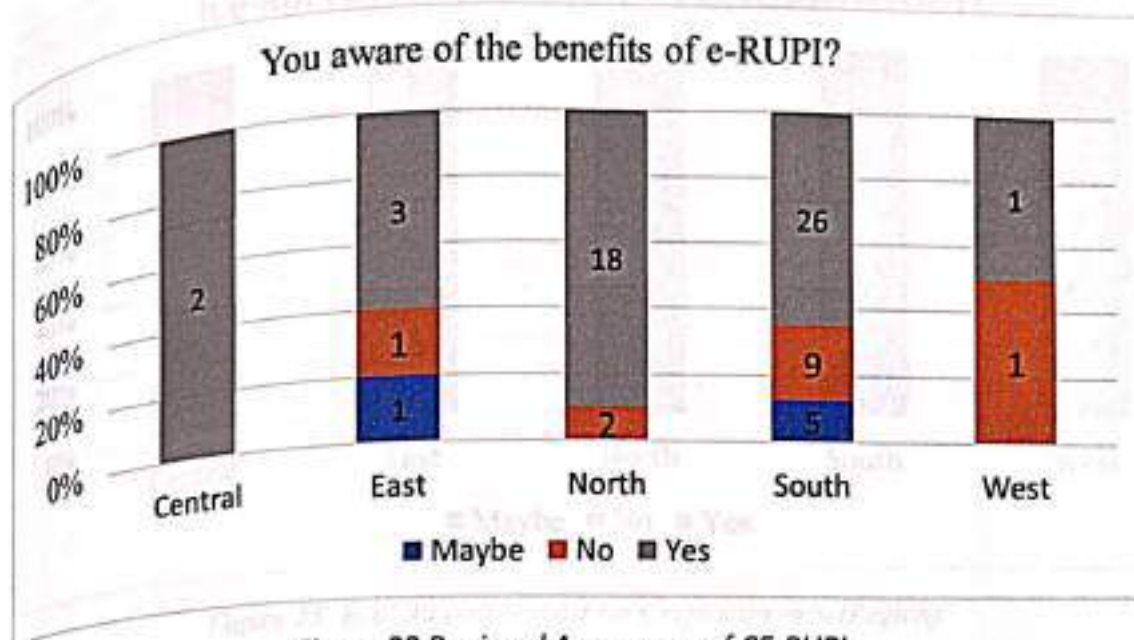


Figure 23 Regional Awareness of CE-RUPI

As depicted in the figure 22 and figure 23, 78% of the females and 69% of the males were aware of the benefits of E-RUPI. In terms of percentage North Region was more aware as compared to any other region having over 90% awareness of benefits of E-RUPI

5.5 Is E-RUPI a Counterpart for Cryptocurrency

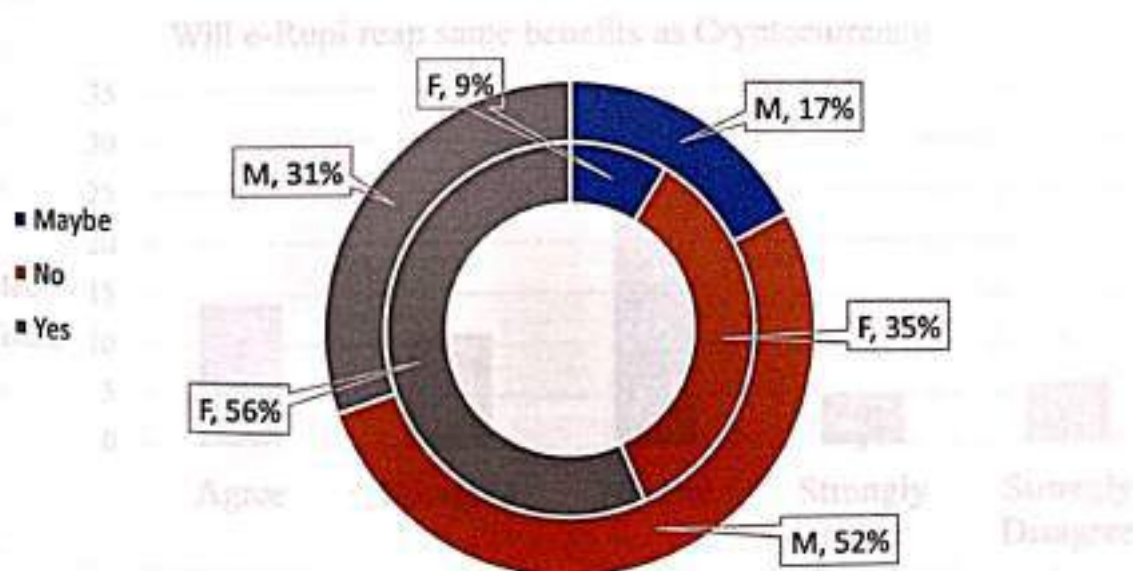


Figure 24 E-RUPI - A counterpart for Cryptocurrency

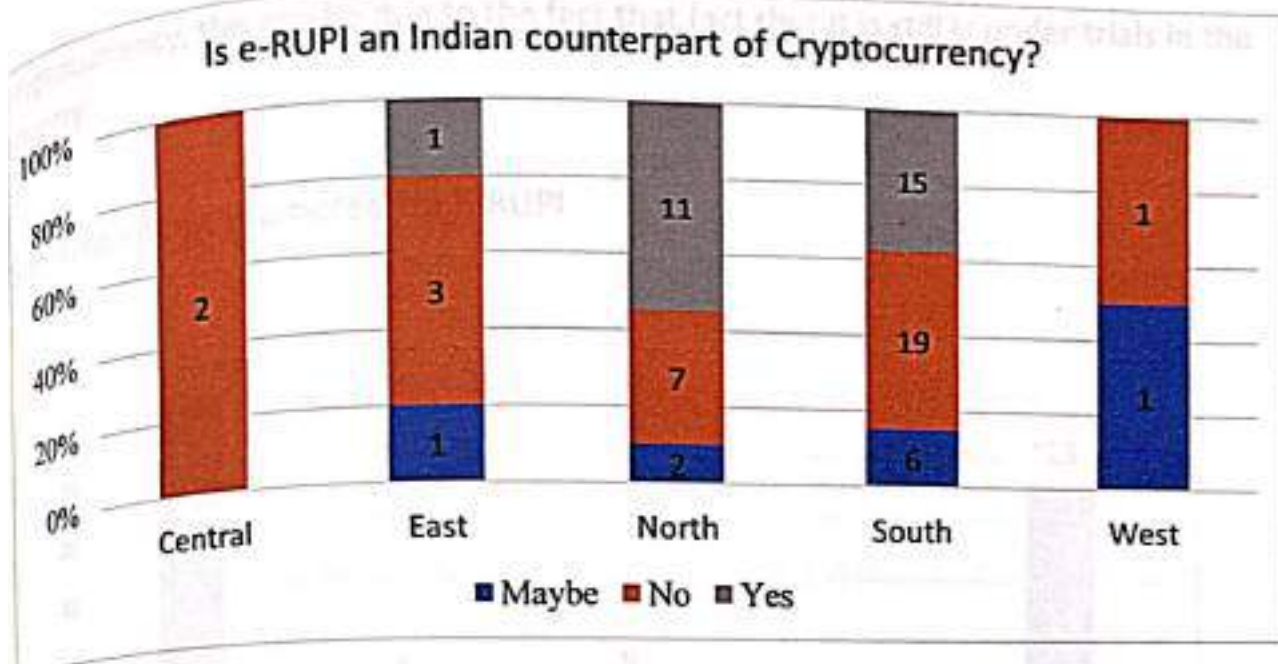


Figure 25 E-RUPI counterpart for Cryptocurrency (Region)

It is observed that 52% males and 56% females do not consider E-RUPI is not a counterpart for cryptocurrency as in Figure 24.

It is observed that the as contrary to the general opinion the of the majority, the North region perceives the E-RUPI to be a counterpart for the cryptocurrency. In north region 55% of people perceive it is a counterpart for cryptocurrency followed by the south region where 33% people consider the same.

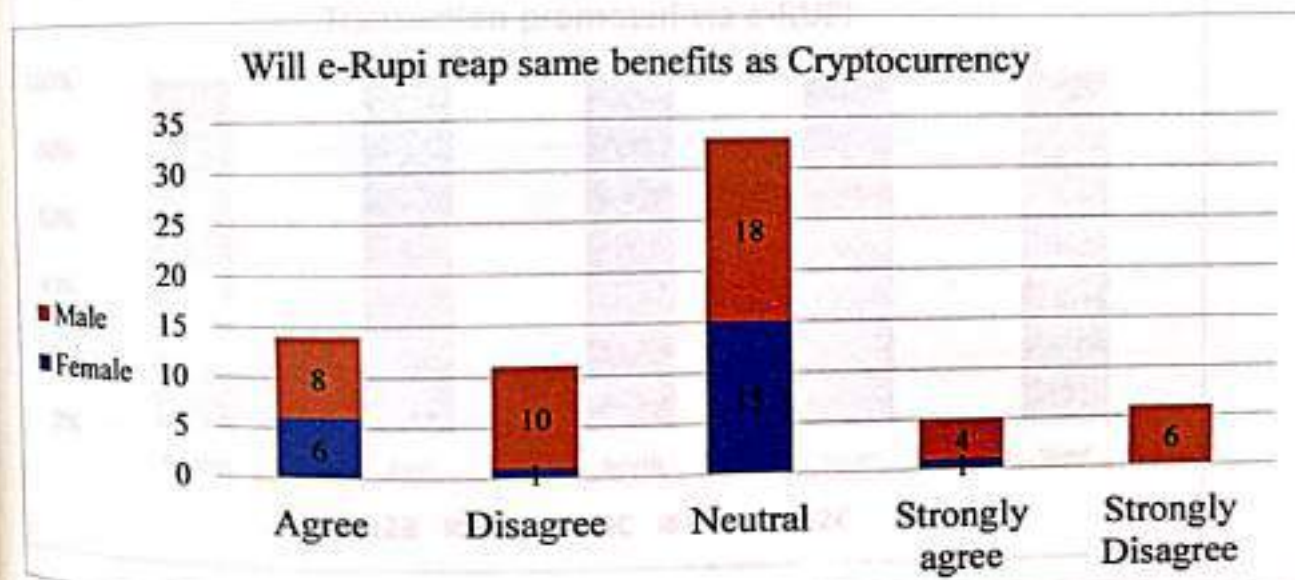


Figure 26 Will E-RUPI reap same benefits as cryptocurrency

It was concluded from the Figure 26 that majority of the stakeholders cannot form an opinion on the statement of E-RUPI having same raping same benefits as

cryptocurrency, this can be due to the fact that it is still under trials in the country

5.6 Transactions promoted via E-RUPI

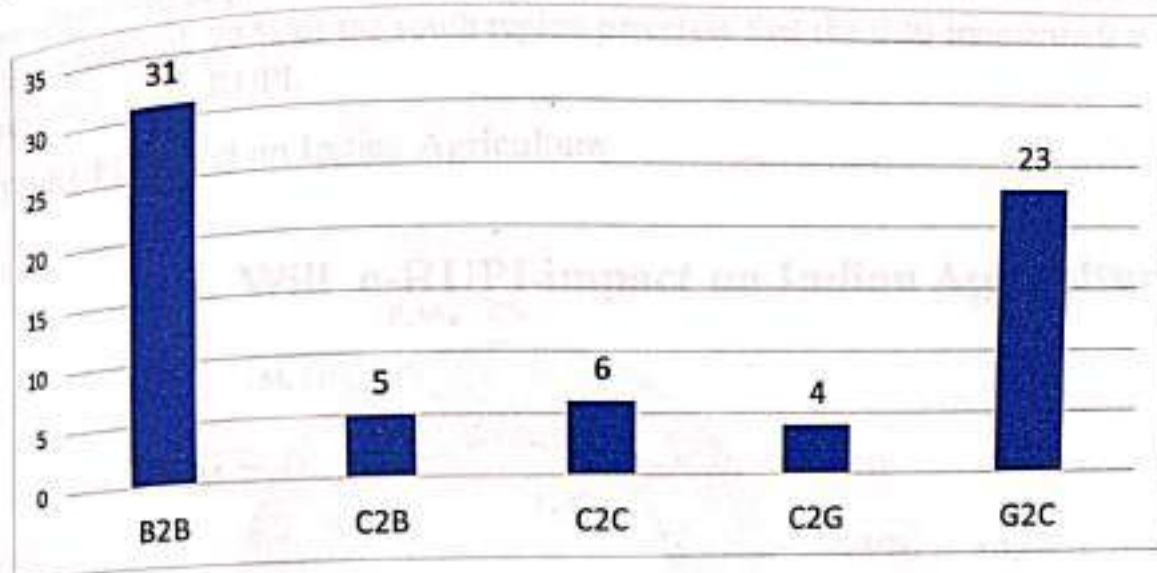


Figure 27 Transaction done through E-RUPI

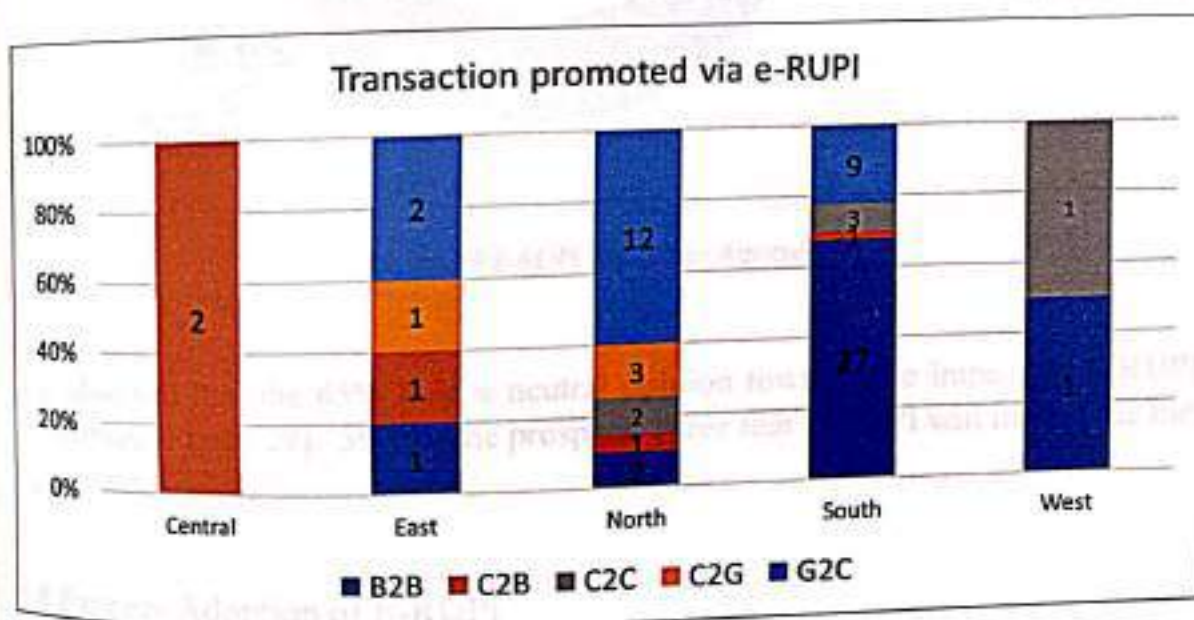


Figure 28 Transactions promoted through E-RUPI (Region)

As observed (Figure 28) Business to Business transactions will be promoted the most followed by Government to consumer.

The retired persons believe that the transactions promoted via e-RUPI are Government to Customer transaction. 47% of the working professionals perceive that G2C transactions will be promoted. The students on the end believe that B2B transaction will benefit the most out of E-RUPI method of transaction contributing 47%

It was observed that the central region perceives that the customer to business transactions will be promoted the most the East region perceives that all the transactions will be promoted. 70 % of the south region perceives that the B2B transactions will be promoted using E-RUPI.

5.7 E-RUPI impact on Indian Agriculture

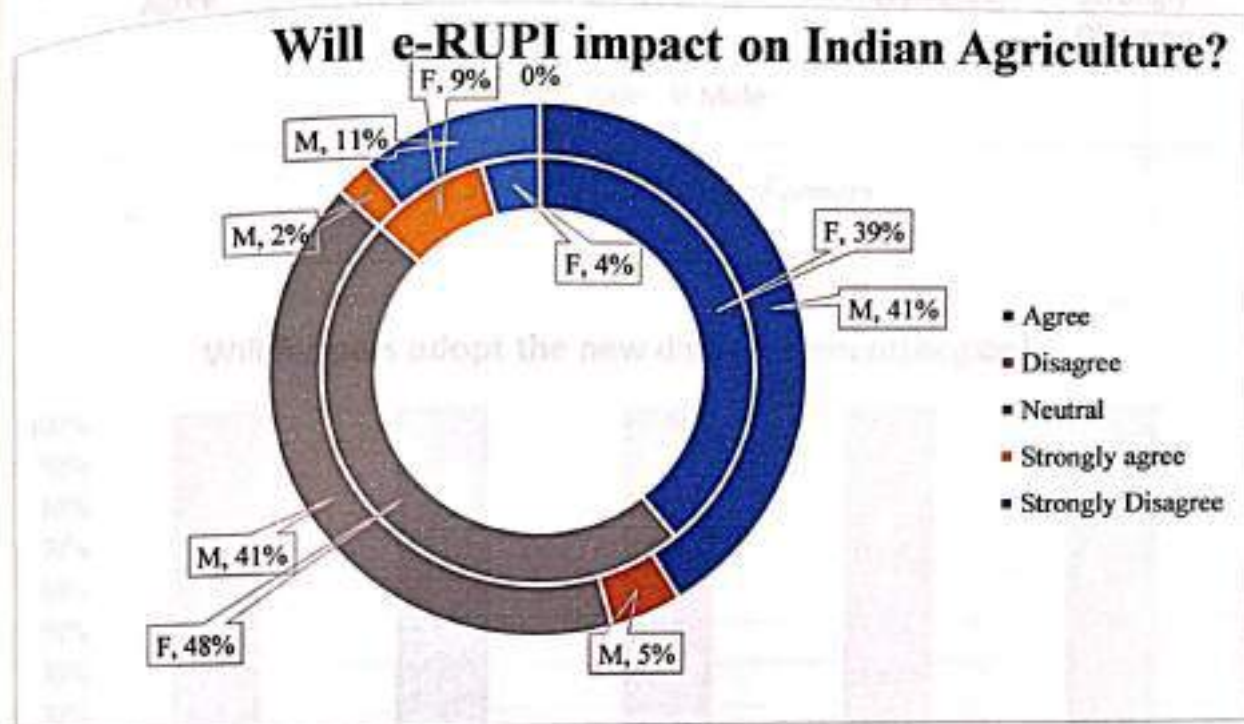


Figure 29 E-RUPI impact on Agriculture

It is observed that the 45% hold a neutral opinion towards the impact of E-RUPI in Agriculture (Figure 29). 39% of the prospects agree that E-RUPI will impact the Indian Agriculture.

Figure 31 Farmers adoption of E-RUPI (Logan)

5.8 Farmers Adoption of E-RUPI

It was observed that the prospects hold a neutral stand for the adoption of E-RUPI by farmers (Figure 31). Figure 31 depicts the adoption of E-RUPI by farmers. The majority perceive that the farmers are likely to be impacted in a positive way. 45% of the prospects from the south region and 10% from North disagree to farmer's adoption of E-RUPI (Figure 31).

Will e-RUPI help the farmers

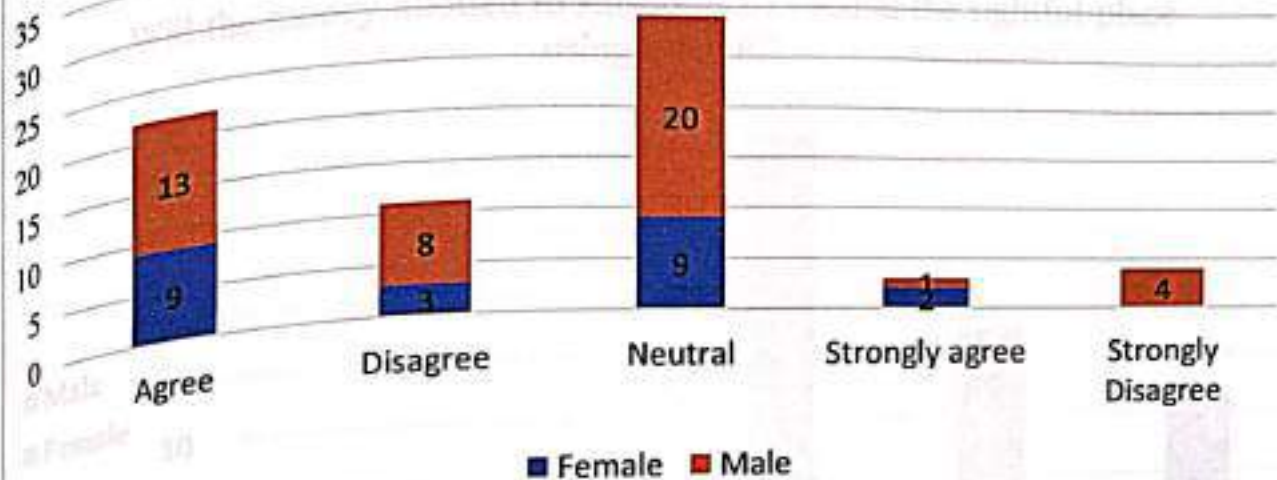


Figure 30 Will E-RUPI help Farmers

Will farmers adopt the new digital payment(Region)

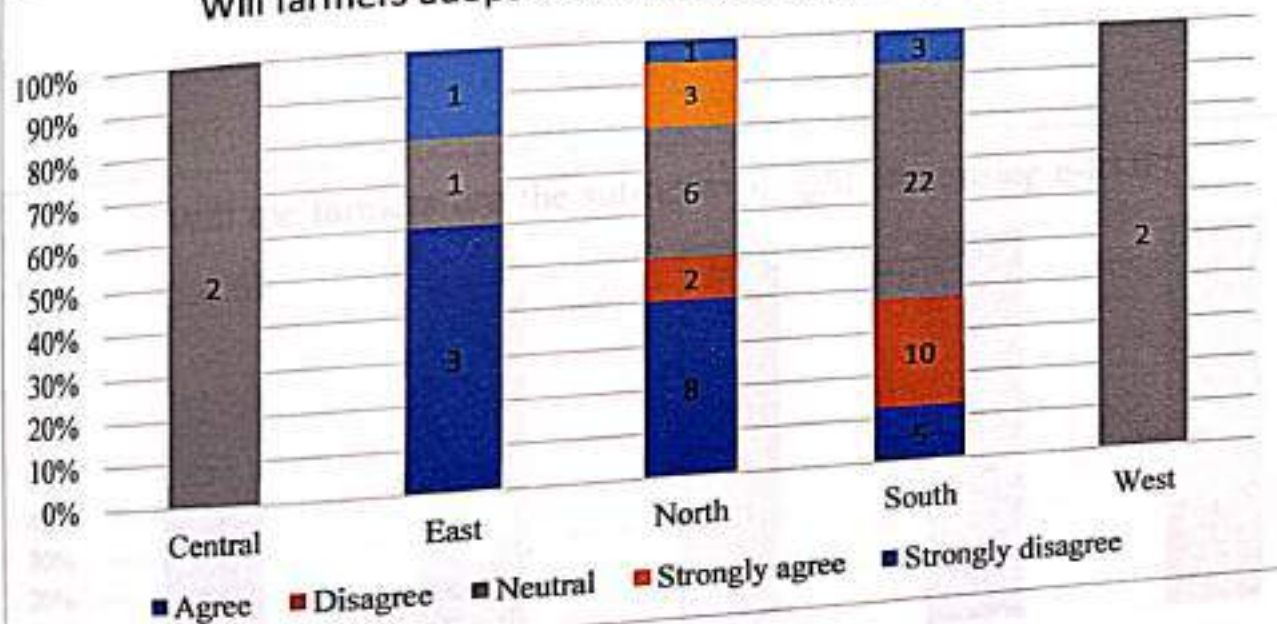


Figure 31 Farmers adoption of E-RUPI(Region)

It was observed that the prospects hold a neutral stand for the adoption of E-RUPI by farmers (Figure 30). Figure 31 depicts 40% off the students and working agri-professionals hold a neutral stand on the impact of E-RUPI. The majority perceive that the farmers are likely to be impacted in a positive way. 30 % of the prospects from the south region and 10% from North disagree to farmer's adoption of E-RUPI (Figure30)

5.9 Rightful Use of Subsidies

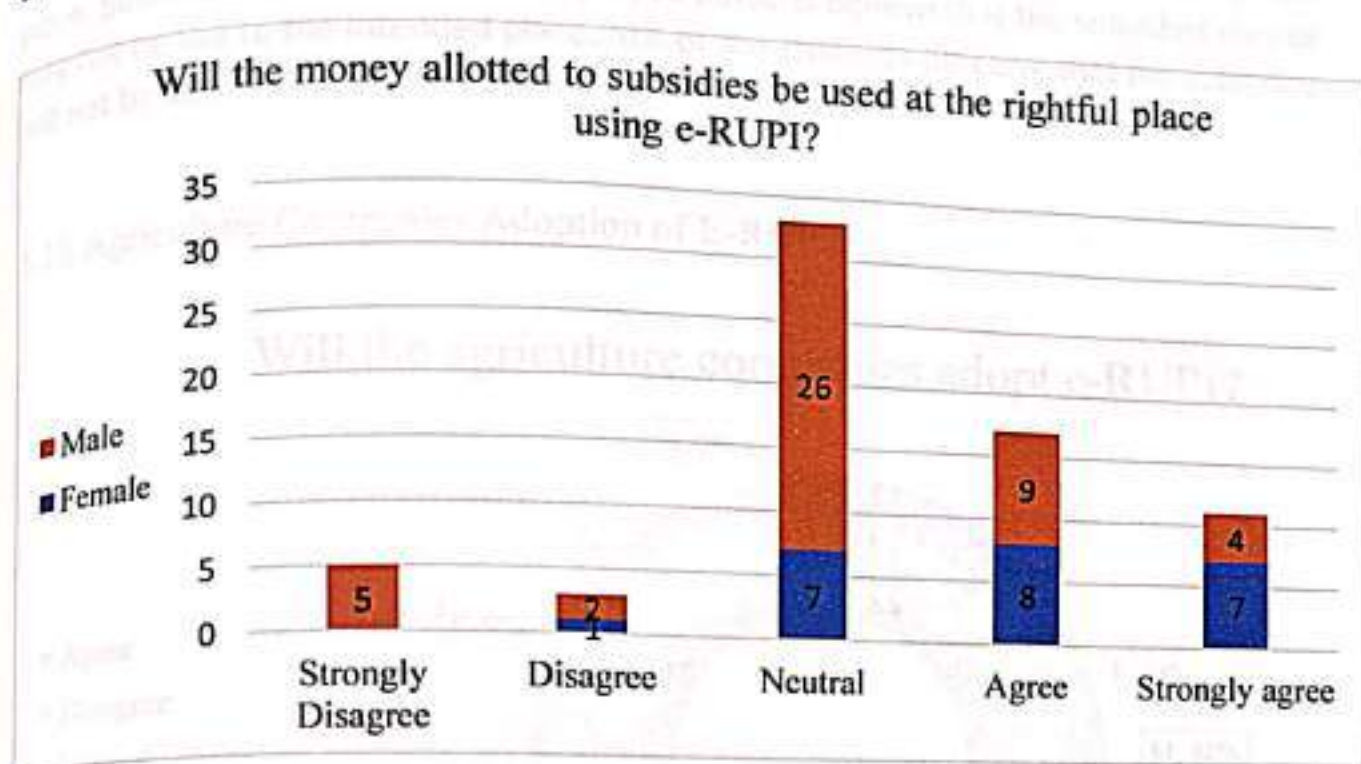


Figure 32 Rightful use of Subsidies

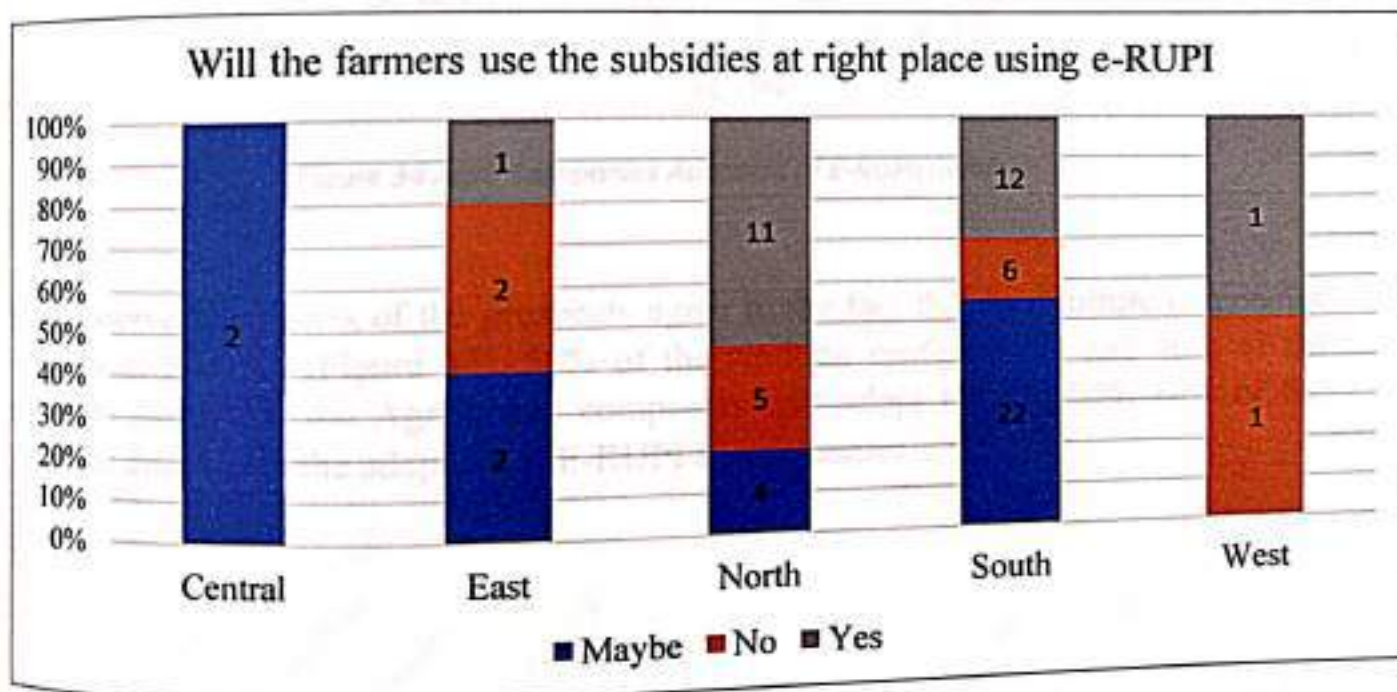


Figure 33 Rightful Use of Subsidies(Region)

The majority of the farmers may or may not use the subsidies at the rightful place. The prospects hold a neutral opinion on the rightful use of the subsidies (Figure32). 50%

of the working professionals perceive that the subsidies will be used at the rightful place. 50% of the students and unemployed persons believe that the subsidies may or may not be use to the intended place. 20% of the students perceive that the subsidies will not be used at the rightful place (Figure 33).

5.10 Agriculture Companies Adoption of E-RUPI

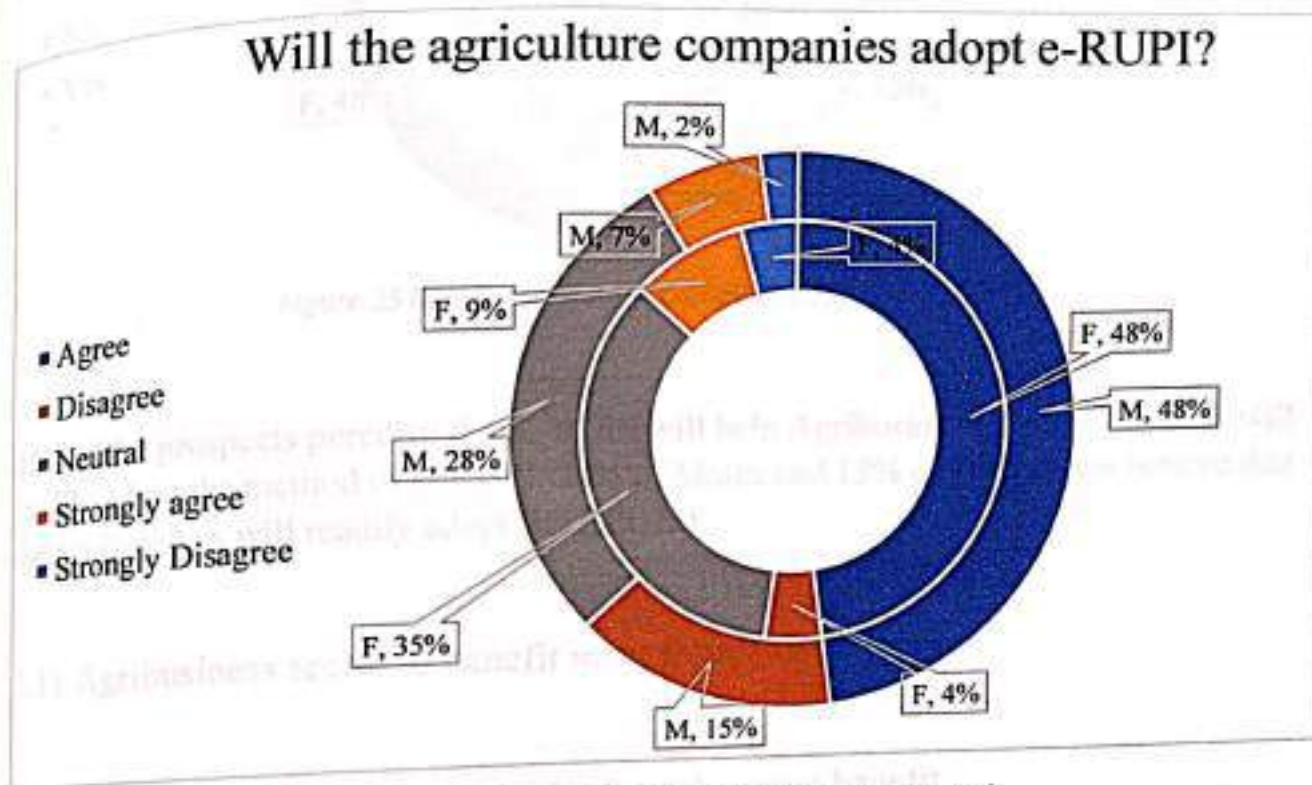


Figure 34 Agri- Companies Adoption of E-RUPI(Gender)

It is observed that 48% of the prospects agree to the fact that Agriculture companies will adopt E-RUPI (Figure 34). 50% of the working professionals and 40% of the students agree that the Agriculture companies will adopt the E-RUPI. 10% of the students disagree to the adoption of E-RUPI by companies

Readiness of Will Agribusiness companies to adopt e-RUPI?

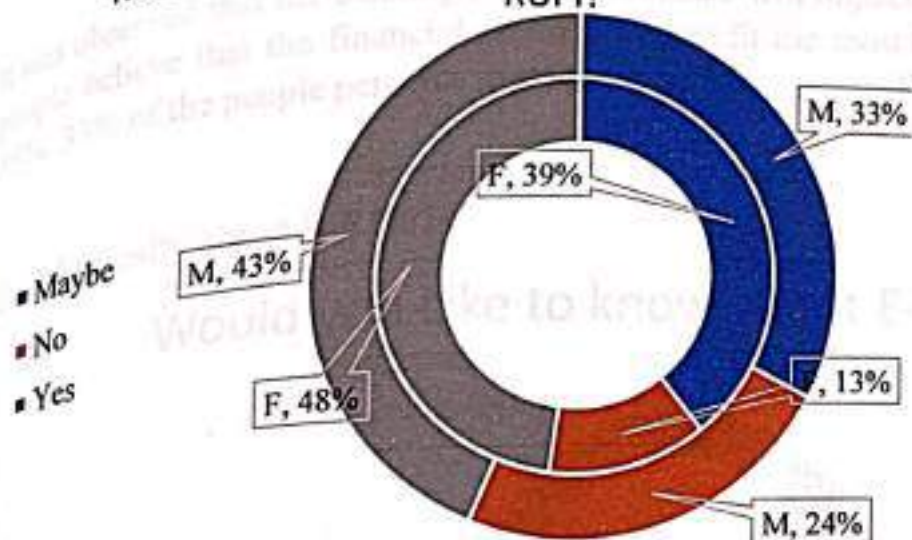


Figure 35 Readiness of Agri- Companies to Adopt E-RUPI

45% pf the prospects perceive that E-RUPI will help Agribusiness and agribusiness will readily adopt the method of E-RUPI. 24% of Males and 13% of the females believe that the agribusiness will readily adopt the E-RUPI.

5.11 Agribusiness sector to benefit most from E-RUPI

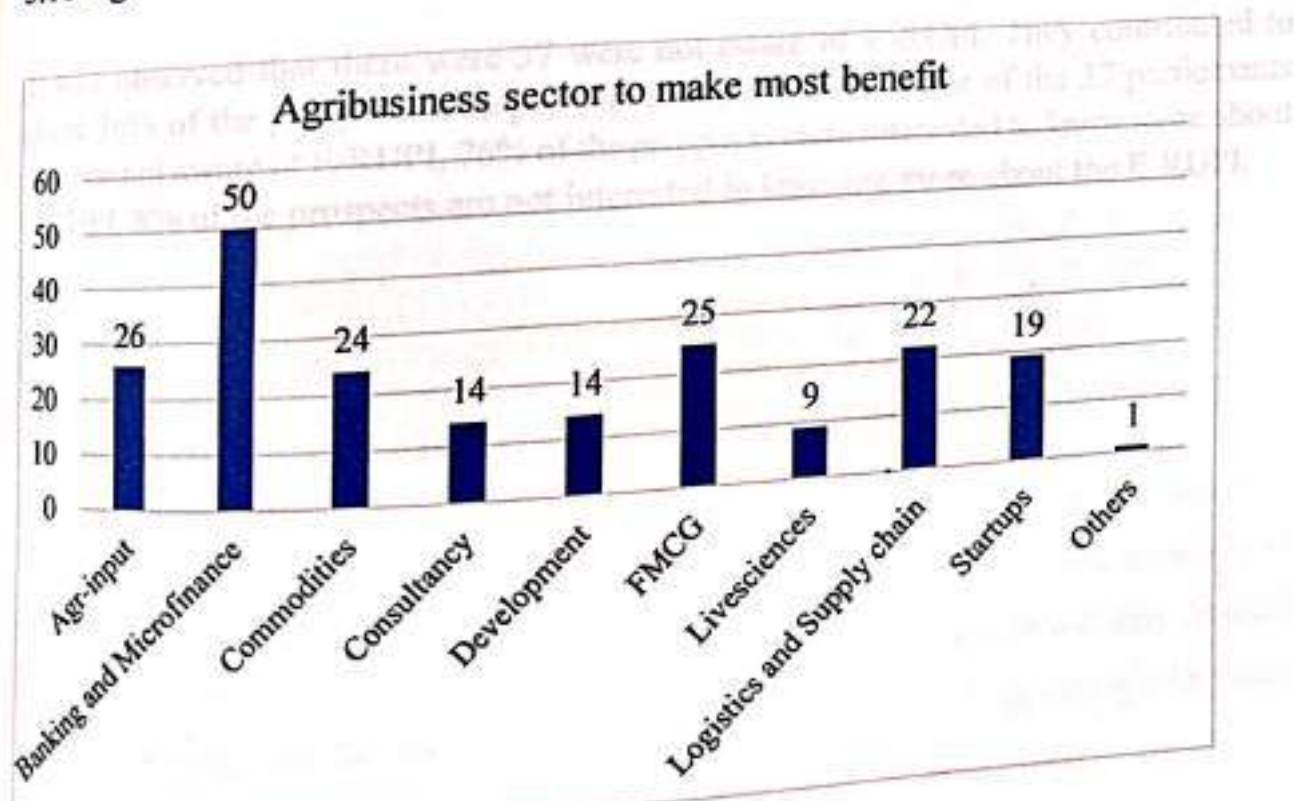


Figure 36 Agri- Sector Benefit most from E-RUPI

Chapter 6: Conclusion and Recommendations

It was observed that the banking and microfinance will impacted the most, 72% of the people believe that the financial sector will benefit the most followed by Agri-input 35%. 33% of the people perceive that FMCG will be impacted.

5.12 Curiosity about E-RUPI

Would you Like to know about E-RUPI

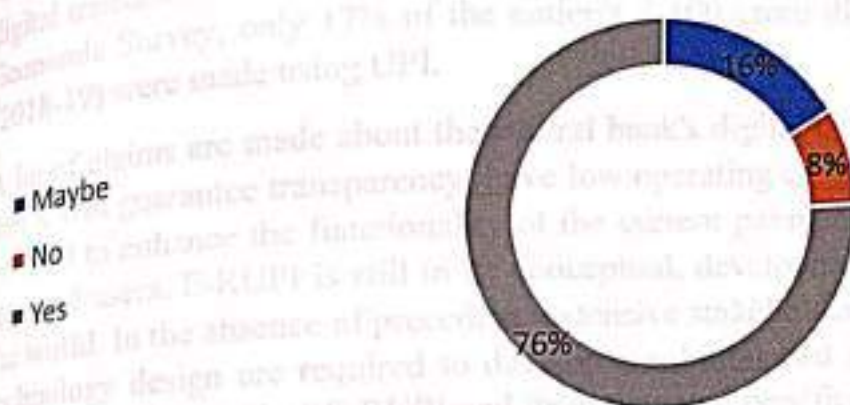


Figure 37 Curiosity about E-RUPI

It was observed that there were 37 were not aware of E-RUPI. They contributed to about 36% of the prospects who participated in the survey. Out of the 37 participants who are not aware of E-RUPI, 76% of the prospects were interested to know more about E-RUPI. 8% of the prospects are not interested in knowing more about the E-RUPI.

Recommendations

- Since it's a new financial technology being introduced in India and currency notes are being carried in Delhi and Mumbai. The people who have accounts in the banks having permission can only use the currency. It's a very limited technology and the awareness about it is very less. Hence spreading awareness about it most important.

Chapter 6: Conclusion and Recommendations

Conclusion

India is one of the progressive countries in terms of bringing innovation in the digital ecosystem of the country specially in digital finance and digital payments. RTGS, NEFT are very one of the successful digital movements. According to the government's pre-Budget Economic Survey, UPI accounted for 52% of the 8,840 crore financial digital transactions with a total value of Rs 126 lakh crore in FY22. According to the Economic Survey, only 17% of the nation's 3,100 crore digital transactions in FY19 (2018-19) were made using UPI.

A lot of claims are made about the central bank's digital currency, E-RUPI, including that it will guarantee transparency, have low operating costs, and have the potential to be used to enhance the functionality of the current payment systems to serve a wider range of users. E-RUPI is still in its conceptual, development, or pilot stages all over the world. In the absence of precedent, extensive stakeholder consultation and iterative technology design are required to develop a solution that satisfies the requirements. Even though the goal of E-RUPI and its anticipated benefits are well understood, it is crucial to find innovative solutions and convincing use cases that will make E-RUPI at least as appealing as money.

Agriculture is a developing sector of the country and always contributes to GDP. From 2011 to 2022, India's GDP from agriculture was 4492.89 billion INR on average, with a record high of 6934.75 billion INR in the fourth quarter of 2022 and a record low of 2690.74 billion INR in the third quarter of 2011. With the inception of the agriculture digitalization such as JAM Trinity and Direct benefit transfer led to the rightful use of the subsidies. With the upcoming technology like E-RUPI in agriculture, the transactions will be simplified, tracked and efficient hence increasing the growth of the sector. The usage of E-RUPI in agriculture will not only help the farmers but also help in the increase the share of the agriculture business companies.

Recommendations

- Since it's a new financial technology being introduced in India and currency trials are being carried in Delhi and Mumbai. The people who have accounts in the banks having permission can only issue the currency. It's a very limited technology and the awareness about it is very less. Hence spreading awareness about it most important.

- As per the survey, the people fear the invasion of the privacy of individual, hence proper monetary and data protection guidelines shall be issued.
- The rural areas shall be equipped with the infrastructure the infrastructure that can manage the transaction.
- Trials shall be conducted for DBT to understand farmers comfort and opinion on it.
- Tie ups with companies working abroad shall be made to promote the international trade and hence increasing the amount of imports by bypassing the SWIFT international system.

References

(PwC CBDC Global Index, 1st Edition, April 2021)

<https://www.pwc.com/gx/en/industries/financial-services/assets/pwc-e-RUPI-global-index-1st-edition-april-2021.pdf>

(CBDC Tracker, n.d.)

<https://www.atlanticcouncil.org/e-RUPItracker/>

. Anton N. Didenko and Ross P. Buckley(2021): "Central Bank Digital Currencies: A potential response to the financial inclusion challenges of the pacific;

<https://www.adb.org/sites/default/files/publication/720016/central-bank-digital-currencies-pacific.pdf>

Nikhil Sridhar and Patrick Horan (2021): "Should Central Banks Offer the Public Token-Based Digital Currencies?";

<https://www.discoursemagazine.com/economics/2021/06/08/should-central-banks-offer-the-public-token-based-digital-currencies/>

. Ahmat, N and S Bashir (2017): "E-RUPI: a monetary policy perspective", Bank Negara Malaysia Staff insights, 2017/11, September

<https://www.bankofengland.co.uk/paper/2020/central-bank-digital-currency-opportunities-challenges-and-design-discussion-paper>

(www.rbi.org.in, 2022)

<https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=19417>

(RBI, 2022)

<https://rbidocs.rbi.org.in/rdocs/PublicationReport/Pdfs/CONCEPTNOTEACB531172E0B4DFC9A6E506C2C24FFB6.PDF>

e-RUPI ,icsi.edi. 2022

https://www.icsi.edu/media/webmodules/e_RUPI.pdf

e-rupi inclusion, PwC, 2022

[e-rupi-an-innovation-to-accelerate-financial-inclusion-in-india.pdf](#)

Annexure

Structures questionnaire

1. Name
2. E-mail
3. Contact number
4. Age
5. Gender
 - a. Male
 - b. Female
 - c. Others
6. Region
 - a. North
 - b. South
 - c. East
 - d. West
 - e. Central
7. Are you related to Agriculture?
 - a. Yes
 - b. No
8. Occupation
 - a. Student
 - b. Retired
 - c. Working Professional
 - d. Unemployed
9. Are you aware about e-RUPI?
 - a. Yes
 - b. No

10. Are you aware about how e-RUPI works?

- a. Yes
- b. No
- c. Maybe

11. Is e-RUPI an Indian counterpart of Cryptocurrency?

- a. Yes
- b. No
- c. Maybe

12. Are you aware of benefits of e-RUPI?

- a. Yes
- b. No

13. Will e-RUPI be able to reap the same benefits as Cryptocurrency?

- a. Yes
- b. No
- c. Maybe

14. E-RUPI will impact the financial system

- a. Positive
- b. Negative
- c. Cant be said at the moment

15. Will e-RUPI help reduction of corruption

- a. Yes
- b. No
- c. Maybe

16. Will subsidies be used at the rightful place?

- a. Yes
- b. No
- c. Maybe

17. Which transactions will be promoted by e-RUPI the most

- a. G2C
- b. B2B
- c. B2C

d. C2B

18. Will e-RUPI impact Indian Agriculture

- a. Yes
- b. No
- c. Maybe

19. As per you, will farmers adopt e-RUPI?

- a. Yes
- b. No
- c. Maybe

20. Will agriculture companies adopt e-RUPI?

- a. Yes
- b. No
- c. Maybe

21. Will agriculture subsidies use at rightful place?

- a. Least likely
- b. Less likely
- c. Can't be said
- d. Likely
- e. Most likely

22. Will agriculture companies be benefitted from the usage of e-RUPI?

- a. Yes
- b. No
- c. Maybe

23. Agribusiness sector who will be benefitted most from e-RUPI

- a. FMCG
- b. Agri-Input
- c. Banking and Microfinance
- d. Commodities and warehousing
- e. Consultancy
- f. Development
- g. Supply chain and logistics

- h. Agri- startups
- i. Lifesciences
- j. Others

24. How will e-RUPI impact the agriculture business from your perspective?

25. What is your opinion on e-RUPI?

26. Suggestions to improve e-RUPI

27. Would you like to know more about e-RUPI?

- a. Yes
- b. No
- c. Maybe

